

Proposed Program Curriculum

BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND ENGINEERING (CSE)

Part-A

(Effective from 2020-21 session)



Department of Computer Science & Engineering
Rajshahi University of Engineering & Technology

1. Title of Academic Program

Bachelor of Science in Computer Science & Engineering

2. Name of the University

Rajshahi University of Engineering & Technology (RUET)

3. Vision of the University

To produce highly-skilled human resources through engineering and technology focused education and research in order to contribute to the socio-economic development of the country.

4. Mission of the University

UM1	To produce technologically-sound professional human resources with high moral and ethical values in order to meet the demand of the society.
UM2	To create a conducive environment to confirm high standard higher education, research and innovation in the area of engineering and technology.
UM3	To create opportunities for multidimensional academic collaboration between universities, industries and professional organizations in higher education, research and innovation.
UM4	To provide and receive on-demand advice and recommendation to solve scientific and technical issues at national and international level.
UM5	To contribute in policy formulation in order to obtain sustainable socio-economic and industrial development of Bangladesh.

5. Name of the Program Offering Entity (Department)

Department of Computer Science & Engineering (CSE)

6. Vision of the Program Offering Entity

To become a nationwide and global center of excellence by empowering innovation, leading technological evolution, and ensuring the advancement, application, and dissemination of knowledge in diverse domains of Computer Science and Engineering via extensive educational programs, industry collaborations, research, and scholarly publications.

7. Mission of the Program Offering Entity

M1	To deliver contemporary educational programs through proper application and continuous dissemination of Computer Science and Engineering knowledge and train students to confront and resolve complex real-world challenges ensuring significant contribution to the prosperity of the nation and society.
M2	To conduct cutting-edge research for the advancement in Computer Science and Engineering and allied fields by collaborating with stakeholders for creating professionals in academia, government and industry.
M3	To create leadership by promoting ethical and virtuous values and prepare students as lifelong learner of enhanced skills and abilities to guide them for industrial revolution.

8. Objectives of the Program Offering Entity

- Assure cutting-edge educational programs by correctly utilizing and sharing computer science and engineering skills.
- Equip students to face and solve complex real-world challenges and ensure their significant contribution to the nation's and society's prosperity.
- Conduct cutting-edge research in the field of computer science and engineering and cooperate with stakeholders to develop experts for various areas.

- d) Foster leadership through the promotion of high ethical and noble principles, as well as equip students with enhanced knowledge and skills as continuous learners to help them navigate the fifth industrial revolution and beyond.

9. Name of the Degree

Bachelor of Science in Computer Science & Engineering abbreviated as B.Sc. Engg. (CSE)

10. Description of the Program

The bachelor degree program requires 160 credits and is extended over a period of 4 years. This program is to be completed in 8 semesters, each year having 2 semesters. Each semester will be of effective 13 weeks duration. Minimum 160 credits and minimum CGPA 2.20 out of 4.00 are required to obtain B.Sc. Engg. degree.

11. Graduate Attributes (based on need assessment)

Learning Outcome Domain	Level Descriptors
Fundamental Skills	F1. demonstrate knowledge and critical understanding of the well-established principles of his/her field of study, and of the way in which those principles have developed; F2. apply underlying concepts and principles outside the context in which they were first studied, including, where appropriate, the application of those principles in an employment context; F3. apply knowledge and skills in addressing issues/solving problems with minimal supervision; F4. evaluate critically the appropriateness of different approaches to solving problems in his/her field of study; F5. support supervision of junior staff via a mentor or a leader/manager; and F6. display advanced digital literacy which is adequate to perform complex tasks and bring about solutions.
Social Skills	S1. communicate and interact effectively and clearly, ideas, information, problems and solutions as a team to peers, experts and non-experts in Bangla and English; S2. express her/himself fluently and spontaneously in English and Bangla; S3. use language flexibly and effectively for social, academic and professional purposes; S4. produce clear, well structured, detailed text on complex subjects, showing controlled use of organizational patterns, connectors and cohesive devices in advanced proficiency level of Bangla and English; S5. demonstrate the ability to incorporate entrepreneurial skills in planning daily activities; and S6. display advanced civic literacy and knowledge, exercising civic rights and obligations at all levels as well as participating in changes for the improvement of Bangladesh society.
Thinking Skills	T1. exercise very substantial degree of autonomy and often significant responsibility in making judgments/decisions towards the management of self, others and for the allocation of substantial resources; and T2. demonstrate professional knowledge and practical skills in both technical and management to lead a team in inexperienced environment.
Personal Skills	P1. engage in self-direction and self-enterprise skills; P2. demonstrate social, professional, environmental and ethical practice/ values; P3. show-case global knowledge and competencies to fulfil employment, entrepreneurial and lifelong learning skills; and P4. contribute significantly to the society).

12. Program Educational Objectives (PEOs)

PEO1	Graduates will function and advance as computer science and engineering professionals in academia, government, and industry by collaborating with stakeholders.
PEO2	Graduates will propose and implement efficient solutions to the complex engineering problems associated with software and hardware by designing, selecting, and utilizing suitable and contemporary tools, skills, and technologies.
PEO3	Graduates will interact effectively and operate as valuable team players by upholding the highest ethical, professional, and social values while preserving the environment.
PEO4	Graduates will procure leadership positions in industry and exercise entrepreneurial skills to inaugurate enterprises by providing innovative technological solutions to national and worldwide concerns.

13. Program Learning Outcomes (PLOs)

PLO1	Engineering knowledge: Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization respectively to the solution of complex engineering problems.
PLO2	Problem analysis: Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PLO3	Design/development of solutions: Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PLO4	Investigation: Conduct investigations of complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PLO5	Modern tool usage: Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations.
PLO6	The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems.
PLO7	Environment and sustainability: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts.
PLO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
PLO9	Individual work and teamwork: Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
PLO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PLO11	Project management and finance: Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PLO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

14. Mapping Mission of the Department with PEOs

Mission Code	PEO1	PEO2	PEO3	PEO4
M1	3	3		2
M2	3	2	3	
M3			3	3

Correlation: 3 – High; 2 – Medium; 1 – Low

15. Mapping PLOs with the PEOs

PEO/PLO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
PEO1	3								3	3	3	3
PEO2	3	3	3	3	3							
PEO3						3	3	3	3			
PEO4					3				3	3	3	3

Correlation: 3 – High; 2 – Medium; 1 – Low

16. Mapping Courses with the PLOs

Course Code	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO10	PLO11	PLO12
CSE 1100	✓							✓				
CSE 1101	✓	✓	✓									✓
CSE 1102	✓	✓	✓									✓
EEE 1151	✓	✓	✓	✓								
EEE 1152	✓	✓	✓		✓							
Math 1113	✓	✓										
Hum 1113	✓								✓	✓		
Hum 1114					✓				✓	✓		✓
Chem 1113	✓			✓								
Chem 1114	✓		✓		✓							
CSE 1200		✓	✓	✓								✓
CSE 1201	✓		✓	✓								
CSE 1202	✓		✓	✓	✓				✓	✓		
CSE 1203	✓		✓	✓					✓			
CSE 1204	✓		✓	✓					✓			
EEE 1251	✓	✓	✓	✓	✓							
EEE 1252			✓	✓	✓							
Math 1213	✓	✓										
Phy 1213	✓	✓										
Phy 1214	✓	✓										
CSE 2100					✓		✓		✓	✓	✓	
CSE 2101	✓	✓	✓									
CSE 2102	✓	✓	✓									
CSE 2103		✓	✓	✓								
CSE 2104			✓				✓					
EEE 2151	✓	✓		✓								
EEE 2152	✓	✓		✓								
Math 2113	✓	✓										
Hum 2113	✓					✓	✓	✓	✓		✓	✓
CSE 2200					✓			✓	✓	✓		
CSE 2201		✓	✓	✓	✓							
CSE 2202		✓	✓	✓	✓							
CSE 2203		✓	✓									
CSE 2204		✓	✓									
CSE 2205	✓	✓			✓							
CSE 2206		✓			✓							
Math 2213	✓	✓										
Hum 2213									✓		✓	
CSE 3100					✓	✓	✓		✓		✓	✓
CSE 3101		✓			✓			✓				
CSE 3102		✓	✓		✓							

Proposed Program Curriculum

**BACHELOR OF SCIENCE IN
COMPUTER SCIENCE AND ENGINEERING (CSE)**

Part-B

(Effective from 2020-21 session)



Department of Computer Science & Engineering
Rajshahi University of Engineering & Technology

17. Structure of the Curriculum

a) Duration of the Program: Years: 4 Semesters: 8

b) Admission Requirements:

1. The four academic years of study for the Bachelor degree have been designated as 1st year class, 2nd year class, 3rd year class and 4th year class in succeeding higher levels of study. For Architecture, five years of study for the Bachelor degree have been designated as 1st year class, 2nd year class, 3rd year class, 4th year class and 5th year class in succeeding higher levels of study. Students shall be admitted into the 1st year class.
2. The Academic Council will form an Admission Committee in each academic session for admission into 1st year Bachelor Degree class.
3. A candidate for admission into the 1st year class must have passed the H.S.C Examination from a Secondary and Higher Secondary Education Board in Bangladesh (after 12 years of schooling) with Physics, Chemistry, Mathematics and English as his/her subjects of Examination in Higher Secondary level or examination recognized as equivalent thereto, and must also fulfill all other requirements as prescribed by the Academic Council on the recommendation of the Admission Committee. In case of confusion regarding the equivalence, the case may be referred to Equivalence Committee.
4. All candidates for admission into the courses of Bachelor Degree must be the citizens of Bangladesh. Candidates for all seats except the reserved (Tribal) ones, if any, are selected on the basis of merit. However, all candidates must pass the required level as set by the admission committee. The Academic Council, on the recommendation of the Admission Committee, frames the rules for admission into the reserved seats.
5. No student ordinarily is admitted in the 1st year class after the corresponding classes start or after the call goes out for admission into the next session, whichever is earlier.
6. Admission of a newly admitted student in the 1st year class is canceled if he/she fails to attend any class within the first two consecutive weeks after the start of class without prior permission. The date of commencement of classes for the newly admitted students will be announced in advance.
7. An Equivalence Committee consisting of at least five members will be formed by the Academic Council in order to consider the equivalence of different public examinations.
8. A candidate, seeking admission on transfer from other University, should apply to the Registrar of the University if there is any exchange program with that university. The Registrar will refer the case to the concerned Head of the Department and also to the Equivalence Committee. On receiving the opinions of the Head of the Department and of the Equivalence Committee, the matter will be forwarded to the Academic Council. The Academic Council's decision will be communicated to the Head of the Department and the candidate.
9. There is no transfer in the 1st year class. In special cases, students may be admitted into a higher class under clause 8.
10. Every student being admitted to the University shall be examined by a competent medical officer as prescribed in the admission rules

c) Total minimum credit requirement to complete the program: According to BNQF (Part B) for Higher Education:

Minimum credit requirements for the award of Bachelor Degree will be recommended by the respective Academic Committee to the Academic Council.

d) Total class weeks in a Year/Semester:

On the approval of the Academic Council an academic schedule for the year will be announced for general notification before the start of the academic year. The schedule may be prepared according to the following guidelines:

1st Semester	Duration
Classes	13 weeks
Mid-semester recess	1 week
Recess before examination and Semester Final Examination	29 days
Inter-Semester Recess	1 week
2nd Semester	Duration
Classes	13 weeks
Mid-semester recess	1 week
Recess before examination and Semester Final Examination	29 days
Inter-Year Recess	1 week
Vacation and others	Rest
Total	52 weeks
Short Semester	Duration
Classes and Examinations	10 Weeks

e) Minimum CGPA requirements for graduation:

The minimum CGPA requirements for obtaining a Bachelor Degree are 2.20.

f) Maximum Academic Years of Completion:

A student must complete his/her studies within a maximum period of seven years for 4-year bachelor degree and eight years for 5-year bachelor degree.

g) Category of Courses:

- General Education Courses: General education courses refer to interdisciplinary courses, beyond B.Sc. in CSE program that provides well-rounded learning experience to the students of an academic program. Mathematics, physics, chemistry and humanities courses such as economics, government, society, industrial management, accountancy and English belong to this type.

BNQF Code	Course Code	Course Title	Theory Hrs. / Week	Sessional Hrs. / Week	Credit
0541	Math 1113	Differential and Integral Calculus	3	0	3.00
0231	Hum 1113	Functional English	3	0	3.00
0232	Hum 1114	Functional English Sessional	0	3/2	0.75
0531	Chem 1113	Inorganic and Physical Chemistry	3	0	3.00

0531	Chem 1114	Inorganic and Physical Chemistry Sessional	0	3/2	0.75
0541	Math 1213	Co-ordinate Geometry and Ordinary Differential Equation	3	0	3.00
0533	Phy 1213	Physics	3	0	3.00
0533	Phy 1214	Physics Sessional	0	3/2	0.75
0541	Math 2113	Vector Analysis and Linear Algebra	3	0	3.00
0311 0312 0314	Hum 2113	Economics, Government and Sociology	3	0	3.00
0541	Math 2213	Complex Variable, Partial Differential Equation and Harmonic Analysis	3	0	3.00
0411 0413	Hum 2213	Industrial Management and Accountancy	3	0	3.00

- ii. Core Courses: Core courses refer to the courses that characterize the discipline, CSE. All the CSE and EEE courses belong to this category.

BNQF Code	Course Code	Course Title	Theory Hrs. / Week	Sessional Hrs. / Week	Credit
0611	CSE 1100	Computer Fundamentals and Ethics	0	3/2	0.75
0613	CSE 1101	Structured Programming	3	0	3.00
0613	CSE 1102	Structured Programming Sessional	0	3	1.50
0713	EEE 1151	Basic Electrical Engineering	3	0	3.00
0713	EEE 1152	Basic Electrical Engineering Sessional	0	3/2	0.75
0613	CSE 1200	Competitive Programming Sessional	0	3/2	0.75
0613	CSE 1201	Data Structure	3	0	3.00
0613	CSE 1202	Data Structure Sessional	0	3	1.50
0613	CSE 1203	Object Oriented Programming	3	0	3.00
0613	CSE 1204	Object Oriented Programming Sessional	0	3	1.50
0714	EEE 1251	Electronic Devices and Circuits	3	0	3.00
0714	EEE 1252	Electronic Devices and Circuits Sessional	0	3	1.50
0541	CSE 2101	Discrete Mathematics	3	0	3.00
0541	CSE 2102	Discrete Mathematics Sessional	0	3	1.50
0611	CSE 2103	Digital Logic Design	3	0	3.00
0611	CSE 2104	Digital Logic Design Sessional	0	3	1.50
0713	EEE 2151	Electrical Drives and Instrumentation	3	0	3.00
0713	EEE 2152	Electrical Drives and Instrumentation Sessional	0	3/2	0.75
0232	CSE 2200	Technical Writing and Presentation Sessional	0	3/2	0.75
0613	CSE 2201	Algorithm Analysis and Design	3	0	3.00
0613	CSE 2202	Algorithm Analysis and Design Sessional	0	3	1.50
0541	CSE 2203	Numerical Methods	3	0	3.00
0541	CSE 2204	Numerical Methods Sessional	0	3/2	0.75
0611	CSE 2205	Microprocessors, Microcontrollers and Assembly Language	3	0	3.00
0611	CSE 2206	Microprocessors, Microcontrollers and Assembly Language Sessional	0	3	1.50
0612	CSE 3101	Database Systems	3	0	3.00
0612	CSE 3102	Database Systems Sessional	0	3	1.50
0611	CSE 3103	Theory of Computation	3	0	3.00
0611	CSE 3105	Computer Interfacing and Embedded Systems	3	0	3.00
0611	CSE 3106	Computer Interfacing and Embedded Systems Sessional	0	3/2	0.75
0611	CSE 3107	Computer Architecture	3	0	3.00

0611	CSE 3108	Computer Architecture Sessional	0	3/2	0.75
0542	CSE 3109	Applied Statistics and Queuing Theory	3	0	3.00
0613	CSE 3201	Operating Systems	3	0	3.00
0613	CSE 3202	Operating Systems Sessional	0	3/2	0.75
0612	CSE 3203	Data Communication	3	0	3.00
0612	CSE 3204	Data Communication Sessional	0	3/2	0.75
0613	CSE 3205	Software Engineering	3	0	3.00
0613	CSE 3206	Software Engineering Sessional	0	3/2	0.75
0613	CSE 3207	Artificial Intelligence	3	0	3.00
0613	CSE 3208	Artificial Intelligence Sessional	0	3/2	0.75
0613	CSE 3209	Digital Signal Processing	3	0	3.00
0613	CSE 3210	Digital Signal Processing Sessional	0	3/2	0.75
0611	CSE 4101	Compiler Design	3	0	3.00
0611	CSE 4102	Compiler Design Sessional	0	3/2	0.75
0612	CSE 4103	Computer Networks	3	0	3.00
0612	CSE 4104	Computer Networks Sessional	0	3	1.50
0613	CSE 4105	Digital Image Processing	3	0	3.00
0613	CSE 4106	Digital Image Processing Sessional	0	3/2	0.75
0613	CSE 4201	Computer Graphics	3	0	3.00
0613	CSE 4202	Computer Graphics Sessional	0	3/2	0.75
0613	CSE 4203	Machine Learning	3	0	3.00
0613	CSE 4204	Machine Learning Sessional	0	3/2	0.75
0611	CSE 4205	Security and Privacy	3	0	3.00
0611	CSE 4206	Security and Privacy Sessional	0	3/2	0.75
0031	CSE 4208	Seminar	0	3/2	0.75

- iii. Elective Courses: Elective courses refer to the courses that are defined for specialization within the discipline, CSE. Elective courses have been introduced in 7th and 8th semesters.

BNQF Code	Course Code	Course Title	Theory Hrs. / Week	Sessional Hrs. / Week	Credit
0413	CSE 4109	Information Systems Analysis and Design	3	0	3.00
0413	CSE 4110	Information Systems Analysis and Design Sessional	0	3/2	0.75
0613	CSE 4111	Unix Programming	3	0	3.00
0613	CSE 4112	Unix Programming Sessional	0	3/2	0.75
0611	CSE 4113	Digital System Design	3	0	3.00
0611	CSE 4114	Digital System Design Sessional	0	3/2	0.75
0613	CSE 4115	Simulation and Modeling	3	0	3.00
0613	CSE 4116	Simulation and Modeling Sessional	0	3/2	0.75
0612	CSE 4117	Wireless Networks	3	0	3.00
0612	CSE 4118	Wireless Networks Sessional	0	3/2	0.75
0613	CSE 4119	Data Mining	3	0	3.00
0613	CSE 4120	Data Mining Sessional	0	3/2	0.75
0613	CSE 4121	Computer Vision	3	0	3.00
0613	CSE 4122	Computer Vision Sessional	0	3/2	0.75
0613	CSE 4123	Knowledge Engineering	3	0	3.00
0613	CSE 4124	Knowledge Engineering Sessional	0	3/2	0.75
0611	CSE 4209	VLSI Design	3	0	3.00
0611	CSE 4211	Parallel and Distributed Processing	3	0	3.00
0922	CSE 4213	Impact of Computer on Society	3	0	3.00
0922	CSE 4215	Decision Support System	3	0	3.00
0612	CSE 4217	Network Planning	3	0	3.00
0922	CSE 4219	Human Computer Interaction	3	0	3.00
0611	CSE 4221	Switching Systems	3	0	3.00

0611	CSE 4223	Control System Engineering	3	0	3.00
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- iv. Capstone Projects/Industrial Attachment/Thesis/Projects: Capstone projects are integrated with most of the sessional courses. Apart from capstone projects, there are two software development projects, one web development project, a two-semester long thesis (7th and 8th semester) and an industrial attachment while conducting the CSE program throughout 8 semesters.

BNQF Code	Course Code	Course Title	Theory Hrs. / Week	Sessional Hrs. / Week	Credit
0613	CSE 2100	Software Development Project I	0	3/2	0.75
0613	CSE 3100	Web Based Application Project	0	3/2	0.75
0613	CSE 3200	Software Development Project II	0	3/2	0.75
061	CSE 4000	Project/Thesis	0	2+6	4.00
0031	CSE 4108	Industrial Attachment	0	3/2	0.75

18. Year/Level/Semester/Term wise distribution of courses

a. Course Distribution Summary

Course Type	Percentage of Credit (%)	Credits
Mathematics and Basic Sciences		
(a) Mathematics	7.50	12.00
(b) Physics	2.34	3.75
(c) Chemistry	2.34	3.75
Humanities		
(a) Economics, Government & Sociology	1.88	3.00
(b) Industrial Management & Accountancy	1.88	3.00
(c) English with Sessional	2.34	3.75
Basic and Major Engineering		
(i) Electrical Engineering with Sessional	7.50	12.00
(ii) Core Engineering		
(a) Theoretical	52.50	84.00
(b) Sessional, Project and Thesis	21.72	34.75
Total	100.00	160.00

b. Summary of Undergraduate Course Plan

Sl. No.	Year/Semester	Theory		Sessional		Total Credits
		No. of Course	Credits	No. of Course	Credits	
1	1 st / 1 st	5	15.00	5	4.50	19.50
2	1 st / 2 nd	5	15.00	5	6.00	21.00
3	2 nd / 1 st	5	15.00	4	4.50	19.50
4	2 nd / 2 nd	5	15.00	4	4.50	19.50
5	3 rd / 1 st	5	15.00	4	3.75	18.75
6	3 rd / 2 nd	5	15.00	6	4.50	19.50
7	4 th / 1 st	5	15.00	7	6.25	21.25
8	4 th / 2 nd	5	15.00	5	6.00	21.00
Total=		40	120.00	40	40.00	160.00

c. 1st YEAR 1st SEMESTER

BNQF Code	Course Code	Course Title	Theory Hrs. / Week	Sessional Hrs. / Week	Credit
0611	CSE 1100	Computer Fundamentals and Ethics Sessional	0	3/2	0.75
0613	CSE 1101	Structured Programming	3	0	3.00
0613	CSE 1102	Structured Programming Sessional	0	3	1.50
0713	EEE 1151	Basic Electrical Engineering	3	0	3.00
0713	EEE 1152	Basic Electrical Engineering Sessional	0	3/2	0.75
0541	Math 1113	Differential and Integral Calculus	3	0	3.00
0231	Hum 1113	Functional English	3	0	3.00
0232	Hum 1114	Functional English Sessional	0	3/2	0.75
0531	Chem 1113	Inorganic and Physical Chemistry	3	0	3.00
0531	Chem 1114	Inorganic and Physical Chemistry Sessional	0	3/2	0.75
Total=			15	9.00	19.50

d. 1st YEAR 2nd SEMESTER

BNQF Code	Course Code	Course Title	Theory Hrs. / Week	Sessional Hrs. / Week	Credit
0613	CSE 1200	Competitive Programming Sessional	0	3/2	0.75
0613	CSE 1201	Data Structure	3	0	3.00
0613	CSE 1202	Data Structure Sessional	0	3	1.50
0613	CSE 1203	Object Oriented Programming	3	0	3.00
0613	CSE 1204	Object Oriented Programming Sessional	0	3	1.50
0714	EEE 1251	Electronic Devices and Circuits	3	0	3.00
0714	EEE 1252	Electronic Devices and Circuits Sessional	0	3	1.50
0541	Math 1213	Co-ordinate Geometry and Ordinary Differential Equation	3	0	3.00
0533	Phy 1213	Physics	3	0	3.00
0533	Phy 1214	Physics Sessional	0	3/2	0.75
Total=			15	12.00	21.00

e. 2nd YEAR 1st SEMESTER

BNQF Code	Course Code	Course Title	Theory Hrs. / Week	Sessional Hrs. / Week	Credit
0613	CSE 2100	Software Development Project I	0	3/2	0.75
0541	CSE 2101	Discrete Mathematics	3	0	3.00
0541	CSE 2102	Discrete Mathematics Sessional	0	3	1.50
0611	CSE 2103	Digital Logic Design	3	0	3.00
0611	CSE 2104	Digital Logic Design Sessional	0	3	1.50
0713	EEE 2151	Electrical Drives and Instrumentation	3	0	3.00
0713	EEE 2152	Electrical Drives and Instrumentation Sessional	0	3/2	0.75
0541	Math 2113	Vector Analysis and Linear Algebra	3	0	3.00
0311 0312 0314	Hum 2113	Economics, Government and Sociology	3	0	3.00
Total=			15	9.00	19.50

f. 2nd YEAR 2nd SEMESTER

BNQF Code	Course Code	Course Title	Theory Hrs. / Week	Sessional Hrs. / Week	Credit
0232	CSE 2200	Technical Writing and Presentation Sessional	0	3/2	0.75
0613	CSE 2201	Algorithm Analysis and Design	3	0	3.00
0613	CSE 2202	Algorithm Analysis and Design Sessional	0	3	1.50
0541	CSE 2203	Numerical Methods	3	0	3.00
0541	CSE 2204	Numerical Methods Sessional	0	3/2	0.75
0611	CSE 2205	Microprocessors, Microcontrollers and Assembly Language	3	0	3.00
0611	CSE 2206	Microprocessors, Microcontrollers and Assembly Language Sessional	0	3	1.50
0541	Math 2213	Complex Variable, Partial Differential Equation and Harmonic Analysis	3	0	3.00
0411 0413	Hum 2213	Industrial Management and Accountancy	3	0	3.00
Total=			15	9.00	19.50

g. 3rd YEAR 1st SEMESTER

BNQF Code	Course Code	Course Title	Theory Hrs. / Week	Sessional Hrs. / Week	Credit
0613	CSE 3100	Web Based Application Project	0	3/2	0.75
0612	CSE 3101	Database Systems	3	0	3.00
0612	CSE 3102	Database Systems Sessional	0	3	1.50
0611	CSE 3103	Theory of Computation	3	0	3.00
0611	CSE 3105	Computer Interfacing and Embedded Systems	3	0	3.00
0611	CSE 3106	Computer Interfacing and Embedded Systems Sessional	0	3/2	0.75
0611	CSE 3107	Computer Architecture	3	0	3.00
0611	CSE 3108	Computer Architecture Sessional	0	3/2	0.75
0542	CSE 3109	Applied Statistics and Queuing Theory	3	0	3.00
Total=			15	7.50	18.75

h. 3rd YEAR 2nd SEMESTER

BNQF Code	Course Code	Course Title	Theory Hrs. / Week	Sessional Hrs. / Week	Credit
0613	CSE 3200	Software Development Project II	0	3/2	0.75
0613	CSE 3201	Operating Systems	3	0	3.00
0613	CSE 3202	Operating Systems Sessional	0	3/2	0.75
0612	CSE 3203	Data Communication	3	0	3.00
0612	CSE 3204	Data Communication Sessional	0	3/2	0.75
0613	CSE 3205	Software Engineering	3	0	3.00
0613	CSE 3206	Software Engineering Sessional	0	3/2	0.75
0613	CSE 3207	Artificial Intelligence	3	0	3.00
0613	CSE 3208	Artificial Intelligence Sessional	0	3/2	0.75
0613	CSE 3209	Digital Signal Processing	3	0	3.00
0613	CSE 3210	Digital Signal Processing Sessional	0	3/2	0.75
Total=			15	9.00	19.50

i. 4th YEAR 1st SEMESTER

BNQF Code	Course Code	Course Title	Theory Hrs. / Week	Sessional Hrs. / Week	Credit
061	CSE 4000	Project/Thesis I	0	2	1.00
0611	CSE 4101	Compiler Design	3	0	3.00
0611	CSE 4102	Compiler Design Sessional	0	3/2	0.75
0612	CSE 4103	Computer Networks	3	0	3.00
0612	CSE 4104	Computer Networks Sessional	0	3	1.50
0613	CSE 4105	Digital Image Processing	3	0	3.00
0613	CSE 4106	Digital Image Processing Sessional	0	3/2	0.75
0031	CSE 4108	Industrial Attachment**	0	3/2	0.75
-	CSE ****	Optional I	3	0	3.00
-	CSE ****	Optional I Sessional	0	3/2	0.75
-	CSE ****	Optional II	3	0	3.00
-	CSE ****	Optional II Sessional	0	3/2	0.75
Total=			15	12.50	21.25

**** Industrial Attachment:** Students will be attached with the industries/service agencies after completing their Third year first semester (during any vacation between the start of Third year second semester and the end Fourth year first semester) to gain practical knowledge.

Optional I/II					
BNQF Code	Course Code	Course Title	Theory Hrs. / Week	Sessional Hrs. / Week	Credit
0413	CSE 4109	Information Systems Analysis and Design	3	0	3.00
0413	CSE 4110	Information Systems Analysis and Design Sessional	0	3/2	0.75
0613	CSE 4111	Unix Programming	3	0	3.00
0613	CSE 4112	Unix Programming Sessional	0	3/2	0.75
0611	CSE 4113	Digital System Design	3	0	3.00
0611	CSE 4114	Digital System Design Sessional	0	3/2	0.75
0613	CSE 4115	Simulation and Modeling	3	0	3.00
0613	CSE 4116	Simulation and Modeling Sessional	0	3/2	0.75
0612	CSE 4117	Wireless Networks	3	0	3.00
0612	CSE 4118	Wireless Networks Sessional	0	3/2	0.75
0613	CSE 4119	Data Mining	3	0	3.00
0613	CSE 4120	Data Mining Sessional	0	3/2	0.75
0613	CSE 4121	Computer Vision	3	0	3.00
0613	CSE 4122	Computer Vision Sessional	0	3/2	0.75
0613	CSE 4123	Knowledge Engineering	3	0	3.00
0613	CSE 4124	Knowledge Engineering Sessional	0	3/2	0.75

j. 4th YEAR 2nd SEMESTER

BNQF Code	Course Code	Course Title	Theory Hrs. / Week	Sessional Hrs. / Week	Credit
061	CSE 4000	Project/Thesis II	0	6	3.00
0613	CSE 4201	Computer Graphics	3	0	3.00
0613	CSE 4202	Computer Graphics Sessional	0	3/2	0.75
0613	CSE 4203	Machine Learning	3	0	3.00
0613	CSE 4204	Machine Learning Sessional	0	3/2	0.75
0611	CSE 4205	Security and Privacy	3	0	3.00
0611	CSE 4206	Security and Privacy Sessional	0	3/2	0.75
0031	CSE 4208	Seminar	0	3/2	0.75
-	CSE ****	Optional III	3	0	3.00

-	CSE ****	Optional IV	3	0	3.00
Total=			15	12.00	21.00

Optional III/IV					
BNQF Code	Course Code	Course Title	Theory Hrs. / Week	Sessional Hrs. / Week	Credit
0611	CSE 4209	VLSI Design	3	0	3.00
0611	CSE 4211	Parallel and Distributed Processing	3	0	3.00
0922	CSE 4213	Impact of Computer on Society	3	0	3.00
0922	CSE 4215	Decision Support System	3	0	3.00
0612	CSE 4217	Network Planning	3	0	3.00
0922	CSE 4219	Human Computer Interaction	3	0	3.00
0611	CSE 4221	Switching Systems	3	0	3.00
0611	CSE 4223	Control System Engineering	3	0	3.00

Proposed Program Curriculum

BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND ENGINEERING (CSE)

Part-C

(Effective from 2020-21 session)



Department of Computer Science & Engineering
Rajshahi University of Engineering & Technology

1st YEAR 1st SEMESTER

Course Code: CSE 1100

Course Title: Computer Fundamentals and Ethics

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: The objective of this course is to help students to understand the fundamental terminologies of Computer Science and Engineering, Hardware, Software and Ethics to prepare them for future courses.

Course Contents:

Computer Science and Engineering as a Discipline: Computer Science, Computer Engineering, Theoretical Computer Science, Central Themes of CSE, Subfields of CSE, Course Map for Four Year Undergrad CSE Degree, Role of Mathematics in CSE, Role of Electrical Engineering in CSE, Diversified Applications of CSE: Artificial Intelligence, Machine Learning, Digital Image Processing, Computer Graphics, Computer Security etc.

Hardware: Types and Generation of Computers, Data Representation, Computer Arithmetic, Processor, Types of Memory, Peripherals, Interfacing, Assembler, Compiler, Interpreter, Levels of Programming Language, Data Communication and Computer Network.

Software: Computer Programming, Data Structure & Algorithms, Database System, Types of Software, Software Licenses, Software Engineering, Familiarization with Various Operating Systems (Windows, Linux, Android, IOS Etc.), Computer Operations: Text Processing (MS-WORD, LibreWriter etc.), Spreadsheet (MS-EXCEL, LibreCalc etc.), Presentation (MS-POWERPOINT, LibreImpress etc.), Browser Software (Chrome, Firefox etc.).

Ethics: Introduction to Ethics and Morality, Ethics for IT Workers and IT Users, Privacy, Intellectual Property, Ethical Decisions in Software Development, Ethics of IT Organizations, Cybercrimes, Cyberattacks and Cybersecurity, Laws against Cybercrimes, Professional Responsibility.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding computer science and engineering as a discipline	✓											
CLO2	Understanding fundamental terminologies related to computer hardware.	✓											
CLO3	Understanding fundamental terminologies related to computer software.	✓											
CLO4	Understanding the importance of ethics in professional life.								✓				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Tutorial • Discussion	• Quiz • Viva • Project

		● Interaction ● Audio/Video ● Lab Manual	● Lab Performance ● Lab Final
CLO2	Cognitive (Understand)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Project ● Lab Performance ● Lab Final
CLO3	Cognitive (Understand)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Project ● Lab Performance ● Lab Final
CLO4	Cognitive (Understand)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Project ● Lab Performance ● Lab Final

Course Code: CSE 1101

Course Title: Structured Programming

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: Computer programming has evolved into one of the most efficient and effective methods of solving mathematical and engineering problems. Students majoring in Computer Science and Engineering must improve their skills in designing efficient algorithms as well as developing and evaluating computer programs. This course is essential for enhancing problem-solving and problem-analysis skills.

Course Contents:

Introduction to Structured Programming: Algorithm, Writing, Debugging and Running Programs using Compiler / Interpreter.

Basic Concepts: Basic I/O, Formatted I/O, Different Data Types and their Range, Operators and their Precedence, Operands and Expression, Expression Evaluation, Conditional Logic, Switch-Case, Character, ASCII Value, Reading and Writing Character, Integer to Character Conversion.

Loop: Basic of Loop, while Loop, for Loops, do-while Loop, Entry Controlled and Exit Controlled Loops, Nested Loop, Formulating Problems using Loops.

Array: Basics of Array, Necessity, Declaration, Initialization, Array Manipulation- Accessing through Indices, Accessing using Loops, One-Dimensional and Multi-Dimensional Arrays.

String: Basic String Operations, Difference between String and Character-Array, String I/O, Array of Strings.

Pointer: Introduction to Pointer, Pointer Operations, Pointers and Array, Array of Pointers.

Functions: Defining & Calling of User-defined Functions, Void Functions with No Parameters, Functions with Return Type and Parameters, Call by Value, Local and Global Variables, Scope of Variables, Built-in Functions, Recursive Functions, Passing Arrays in a Function as Parameter, Call by Reference.

Custom data types: Structures, Unions, Enumerations.

File: Basic Files Operations, Opening, Closing and Updating Binary and Sequential Files, File I/O (Read from File and Write in File).

Advanced Topics: Operations on Bits, Preprocessors and Macros.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding the basic principles and concepts of structured programming.	✓											
CLO2	Analyzing decision making and looping concepts in solving problems.		✓										
CLO3	Exploring usage of arrays, strings and pointers.	✓											
CLO4	Applying concepts of functions, custom data types, files and advanced techniques in programming.		✓										
CLO5	Applying fundamental concepts of programming in solving mathematical, engineering and programming problems.			✓									✓

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO2	Cognitive (Analyze)	• Lecture • Tutorial • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO3	Cognitive (Analyze)	• Lecture • Tutorial • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO4	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO5	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction	• Class Test • Final Exam • Assignment

Course Code: CSE 1102
Course Title: Structured Programming Sessional

Contact Hours/Week: 3 Hours
Credit Hour: 1.50

Rationale: Computer programming has evolved into one of the most efficient and effective methods of solving mathematical and engineering problems. Students majoring in Computer Science and Engineering must improve their skills in designing efficient algorithms as well as developing and evaluating computer programs. This course is essential for enhancing problem-solving and problem-analysis skills.

Course Contents:

Sessional on Structured Programming: Introduction, Basics Concepts, Selection Statements, Control Statements, Arrays, Strings, Pointer, Functions, Custom Data Types, Files, Advanced Topics.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding the basic principles and concepts of structured programming.	✓											
CLO2	Analyzing decision making and looping concepts in solving problems.		✓										
CLO3	Exploring usage of arrays, strings and pointers.	✓											
CLO4	Applying concepts of functions, custom data types, files and advanced techniques in programming.		✓										
CLO5	Applying fundamental concepts of programming in solving mathematical, engineering and programming problems.			✓									✓

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Final • Assignment • Lab Task • Project

CLO2	Cognitive (Analyze)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Final • Assignment • Lab Task • Project
CLO3	Cognitive (Analyze)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Final • Assignment • Lab Task • Project
CLO4	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Final • Assignment • Lab Task • Project
CLO5	Cognitive (Apply)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Final • Assignment • Lab Task • Project

Course Code: EEE 1151

Course Title: Basic Electrical Engineering

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: This course is designed to teach about fundamental concepts and solution techniques of DC circuit and AC circuit. The course will help to design electrical circuits to solve real world problems.

Course Contents:

Direct Current: Voltage, Current, Resistance and Power, Laws of Electrical Circuits, Methods of Network Analysis, Capacitance, Types of Capacitors, Capacitors in Series and Parallel, Inductance, Types of Inductors, Faraday's Law and Lenz's Law, Inductors in Series and Parallel.

Alternating Current: Signal and Waveforms, Instantaneous and RMS Values of Current, Voltage and Power, Average Power, AC Analysis for Various Combination of R, L and C Circuits, Phasor Representation of Sinusoidal Quantities, Resonance, Frequency Response, Passive Filters.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Analyze the fundamental concept of electrical circuit elements and electrical parameters.	✓	✓										
CLO2	Apply different circuit theorems and circuit analysis techniques to solve DC circuits.		✓	✓	✓								
CLO3	Apply different circuit theorems		✓	✓	✓								

	and circuit analysis techniques to solve AC circuits.												
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Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Analyze)	- Lecture - Discussion - Interaction	- Class Test - Final Exam
CLO2	Cognitive (Apply)	- Lecture - Discussion - Interaction	- Class Test - Final Exam
CLO3	Cognitive (Apply)	- Lecture - Discussion - Interaction	- Class Test - Final Exam

Course Code: EEE 1152

Course Title: Basic Electrical Engineering Sessional

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: This course is designed to perform the experiments to verify the theoretical knowledge of EEE 1151 course. The course will develop the practical skill, which is essential in designing electrical circuits to solve real-world problems.

Course Contents:

Sessional on Basic Electrical Engineering: Basic of Electrical Circuit, Experiments on Circuit Theorems, Circuit Simulation and Design.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Apply the knowledge of electrical instruments for measuring various circuit parameters.	✓											
CLO2	Analyze the electrical circuit using circuit theorems.		✓										
CLO3	Design the electrical circuit and simulate that circuit.			✓		✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Apply)	- Lecture - Discussion - Interaction - Lab Manual	- Quiz - Viva - Lab Report - Project - Lab Final

	level of productive skills (speaking-writing) in different rhetorical settings.												
CLO2	Use and apply modern strategies, tools and apps to improve listening skill.					✓							✓
CLO3	Apply contextual, academic, and scholarly vocabularies in the pertinent skills.										✓		
CLO4	Response integrated skills properly: writing/speaking response from listening-reading materials.										✓		
CLO5	Deliver academic presentations in different settings.									✓	✓		✓

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Psychomotor (Articulate)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Quiz • Viva • Lab Report • Lab Final
CLO2	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Quiz • Viva • Lab Final
CLO3	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Quiz • Viva • Lab Report • Lab Final
CLO4	Affective (Response)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Quiz • Viva • Lab Report • Lab Final
CLO5	Psychomotor (Embody)	• Lecture • Discussion • Interaction • Audio/Video	• Viva • Lab Report • Lab Final • Project • Presentation

Course Code: Chem 1113
Course Title: Inorganic and Physical Chemistry

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: Study of chemistry involves some valuable topics in which the instructor designs those for the required direction, instruction and explanation. Acquiring skills and knowledge on basic chemistry are

necessary for addressing numerous problems in daily life and having capability to deal with those issues. Chemistry has become an interdisciplinary science and today no scientific work can escape chemistry. To be more precise, the knowledge of chemistry is being particularly used in the domain of computer science and engineering in order to deal with several issues. With all the importance in modern civilization, Chemistry has earned immense attention in present education.

Course Contents:

Chemical Bond: Ionic Bond, Covalent Bond, Co-ordination Bond, Metallic Bond, Hydrogen Bond, Dipole Bond, Vander Waal's Forces, Hybridization, Resonance, Valence Bond Theory (VBT), Molecular Orbital Theory (MOT), Linear Combination of Atomic Orbital (LCAO) Method.

Thermo-chemistry: Types of Energy, Enthalpy of Reaction, Heat of Combustion, Heat of Formation and Heat of Neutralization, Experimental Determination of Thermal Changes during Chemical Reaction.

Solution: Types of Solution, Factors Influencing the Solubility of Substance, Mechanism of Dissolution, Solution of Gases in Liquids, Different Units of Concentration, Distribution Law and its Application, Properties of Dilute Solution, Raoult's Law - its Application, Elevation of Boiling Point, Depression of Freezing Point and Osmotic Pressure.

Conductivity of Electrolytic Solution: Type of Conductors, Conductance, Specific Conductance, Equivalent Conductance, Mechanism of Electrolytic Conductance, Factors Influencing Conductivity, Arrhenius Theory, Law of Independent Migration of Ions and its Applications, Determination of Transport Number, Abnormal Conductance.

Electromotive Forces: Electrochemical Cell, Cell Reaction, Cell Potential, Cell Representation, Measurement of EMF of a Cell, Relation Between EMF and Free Energy, Electrode Potential, Electrochemical Series, Nernst's Equation, Different Types of Reference Electrodes and pH Measurement, Over Potential, Lithium Ion Battery, Fuel Cell its Latest Development.

Photochemistry: Photochemical Reactions, Laws of Photochemistry, Quantum Yield and its Determination, Photosensitized Reaction, Photo-Physical Processes.

Spectroscopy: Quantization of Energy, Basic Elements of Spectroscopy.

Chemistry of Polymerization: Classification, Bonding in Polymer, Thermosetting and Thermoplastic Polymer, Synthesis, Properties and Uses of Some Polymers-Polyethylene, PVC, Bakelite, and Melamine etc.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understand chemical bond and Thermo-chemistry.	✓											
CLO2	Understand the solution and its conductivity.	✓											
CLO3	Analyze photochemistry, spectroscopy and polymerization.				✓								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	- Lecture - Tutorial - Discussion	- Class Test - Final Exam - Assignment

		- Interaction - Audio/Video	Presentation
CLO2	Cognitive (Understand)	- Lecture - Tutorial - Discussion - Interaction - Audio/Video	- Class Test - Final Exam - Assignment - Presentation
CLO3	Cognitive (Analyze)	- Lecture - Tutorial - Discussion - Interaction - Audio/Video	- Class Test - Final Exam - Assignment - Presentation

Course Code: Chem 1114

Contact Hours/Week: 3/2 Hours

Course Title: Inorganic and Physical Chemistry Sessional

Credit Hour: 0.75

Rationale: Chemistry sessional consists of various lab work which will strengthen the learning process of the students. Since chemistry sessional has been designed on the theory course, students will be able to extend their thinking level on chemistry by pursuing hard working in chemistry lab.

Course Contents:

Sessional on Inorganic and Physical Chemistry: Introduction to Lab Experiments, Conducting Lab Experiments, Preparing Lab Report and Presentation

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understand the current chemical and scientific theories including those in Analytical, Inorganic, Organic and Physical Chemistries by extensive laboratory work.	✓											
CLO2	Analyze and carry out scientific experiments as well as accurately record and analyze the results of such experiments.			✓									
CLO3	Create lab reports and presentations by applying appropriate techniques, resources, and modern engineering and IT tools.					✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	Lecture	Quiz

		• Tutorial • Discussion • Interaction • Lab Manual	• Viva • Lab Report • Lab Final
CLO2	Cognitive (Analyze)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Lab Final
CLO3	Cognitive (Create)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Lab Final • Presentation

1st YEAR 2nd SEMESTER

Course Code: CSE 1200

Course Title: Competitive Programming Sessional

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: The competitive programming lab focuses on developing students' problem-solving skills in order to prepare them to master advanced topics such as data structures, algorithms, and so on. Another purpose of this lab is to encourage students to participate in competitive programming, which is a mental sport that takes place via the Internet or a local network and involves participants attempting to program according to set requirements.

Course Contents:

Sessional on Competitive Programming:

Warmup: Basic Input/Output and Conditionals, Basic Mathematics and Introduction to Number Theory, Basic Geometry, Introduction to Different Online Judges.

Loop and Array: Loops and Nested Loops, Array Operations and Techniques i.e., Cumulative Sum, Longest Consecutive Elements, Finding Duplicates etc.

Functions and Strings: Function Building, Specialized Functions, Basic String Operations and Techniques, Substrings, Subsequences, Palindrome.

Searching and Sorting: Searching Techniques i.e., Linear Search, Binary Search, etc., Sorting Techniques i.e., Bubble Sort, Selection Sort, etc.

Advanced Topics: Recursion, Backtracking, Matrix Operations etc.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding and analyzing problem statements and constraints depending on course materials.		✓										
CLO2	Designing workable solutions based on the topics covered that satisfy the requirements.			✓									
CLO3	Investigating and finding the best solution from the set of available solutions.				✓								
CLO4	Getting familiar with, practicing, and solving competitive programming challenges on online judges.												✓

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand & Analyze)	• Lecture • Tutorial • Discussion	• Lab Performance • Viva • Lab Report

		● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Project ● Lab Final
CLO2	Cognitive (Create)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Lab Performance ● Viva ● Lab Report ● Contest ● Quiz ● Project ● Lab Final
CLO3	Cognitive (Evaluate)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Lab Performance ● Viva ● Lab Report ● Quiz ● Project ● Lab Final
CLO4	Psychomotor (Embody)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Lab Performance ● Viva ● Lab Report ● Contest ● Quiz ● Project ● Lab Final

Course Code: CSE 1201
Course Title: Data Structure

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: One of the concentration requirements for the students majoring in Computer Science and Engineering or Electrical and Computer Engineering or Electrical and Electronic Engineering as core course, and an optional course for the students majoring in Electronic and Telecommunication Engineering to develop the skills in implementation, understanding & analyzing and designing data structures for various real life applications. Besides, students are able to assess how the choice of data structures and algorithm design methods impacts the performance of programs.

Course Contents:

Introduction: Concepts and Examples of Elementary Data Objects, Necessity of Structured Data, Types of Data Structure, Ideas on Linear and Nonlinear Data Structure.

Linear Array: Linear Array & its Representation in Memory, Traversing LA, Insertion & Deletion in LA, Multidimensional Array & its Representation in Memory, Algebra of Matrices, Sparse Matrices.

Stack: Stack Representation & Applications; PUSH and POP Operation on Stack. Polish Notation, Reverse Polish Notation; Evaluation of a Postfix Expression; Transforming Infix Expression into Postfix Expression.

Queue: Its Representation, Insertion & Deletion in Queue, Priority Queues, Recursion: [Factorial Function, Fibonacci Sequence, Ackermann Function, Towers of Hanoi].

Linked List: Linked List & its Representation in Memory, Traversing, Searching, Insertion & Deletion Operation on Linked List, Circular List, Header Linked Lists, Two Way Lists.

Complexity Analysis: Algorithm and Flowchart, Asymptotic Notations: Best case, Worst Case, Average Case, Complexity Analysis of Different Algorithms.

Sorting and Searching: Insertion Sort, Selection Sort, Bubble Sort, Quick Sort, Merge Sort, Binary and Linear Search, Hash Function, Collision Resolution.

Tree: Tree Terminology, Representation of Binary Trees in Memory, Traversing Binary Tree, Binary Search Tree, Insertion & Deletion on Binary Search Tree, Balanced Binary Search Tree, AVL tree, Red Black Tree, Insertion & Deletion on Heap, Heap Sort, B Trees, General Tree.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding the concept of data structures and analyzing the complexity of algorithms for different data structures.	✓			✓								
CLO2	Understanding the function of Linear Array and Linked List and applying them in different problems.	✓		✓									
CLO3	Understanding the function of Stack, Queue & Recursion and applying them in different problems.	✓		✓									
CLO4	Implementing and analyzing different sorting and searching techniques.	✓		✓	✓								
CL05	Understanding the function and use of Tree data structure.	✓		✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand & Analyze)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO2	Cognitive (Understand & Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO3	Cognitive (Understand & Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO4	Cognitive (Apply & Analyze)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO5	Cognitive (Understand & Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment

Course Code: CSE 1202
Course Title: Data Structure Sessional

Contact Hours/Week: 3 Hours
Credit Hour: 1.50

Rationale: One of the concentration requirements for the students majoring in Computer Science and Engineering or Electrical and Computer Engineering or Electrical and Electronic Engineering as core course,

and an optional course for the students majoring in Electronic and Telecommunication Engineering to develop the skills in implementation, understanding & analyzing and designing data structures for various real life applications. Besides, students are able to assess how the choice of data structures and algorithm design methods impacts the performance of programs.

Course Contents:

Sessional on Data Structure: Linear Array, Stack, Queue, Recursion, Linked List, Sorting & Searching, Tree.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Solving problems related to Linear Array	✓			✓								
CLO2	Implementing Linked List in different problems	✓		✓									
CLO3	Implementing Stack, Queue & Recursion on real life problems	✓		✓									
CLO4	Understanding the function and implementation of Tree data structure	✓		✓									
CLO5	Implementing and analyzing different sorting and searching algorithms	✓		✓	✓								
CLO6	Design and build innovative software solutions using different data structures for real life problems					✓				✓	✓		

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO2	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO3	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO4	Cognitive (Understand & Apply)	• Lecture • Tutorial • Discussion	• Quiz • Viva • Lab Report

		- Interaction - Audio/Video - Lab Manual	- Project - Lab Final
CLO5	Cognitive (Apply & Analyze)	- Lecture - Tutorial - Discussion - Interaction - Audio/Video - Lab Manual	- Quiz - Viva - Lab Report - Project - Lab Final
CL06	Cognitive (Create)	- Discussion - Interaction	- Lab Report - Project - Presentation - Lab Final

Course Code: CSE 1203

Course Title: Object Oriented Programming

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: Object-oriented programming provides a higher-level way for programmers to envision and develop their applications. It provides insights into some of the rudimentary building blocks of software design and development in a module-oriented fashion. As a Computer Science and Engineering major it is essential to learn the concept of object oriented programming in order to provide professional services to the stakeholders including individuals, offices and community.

Course Contents:

Fundamentals of OOP: Introduction to Object Oriented Programming, Principles of Object Oriented Design, Encapsulation and Information-hiding, Data Binding, Static and Dynamic Binding.

Classes and Objects: Structure of Class, Access Modifiers, Nested Classes, Abstract Classes, Arrays of Objects, Pointer to Objects, Friend Function, Data Abstraction, Static Variable and Function.

Constructors and Destructors: Default Constructor, Copy Constructor, Dynamic Constructor, Deep Copy and Shallow Copy, Constructor Function for Derived Class and their Order of Execution, Destructor.

Inheritance: Single Inheritance vs. Multiple Inheritance, Mode of Inheritance, Virtual Inheritance, Interface.

Polymorphism: Operator and Function Overloading, Run-Time and Compile Time Polymorphism, Virtual Function, Errors and Exception Handling.

Advanced Topics: Persistent Objects, Objects and Portable Data, UML Basics, Namespace, Package, Templates, JFrame, Multithreading, Concept of MVC Framework.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understand fundamental concepts of object-oriented programming.	✓											
CLO2	Apply object-oriented programming concepts while writing programs.	✓		✓									
CLO3	Formulize advanced object-oriented solutions			✓	✓					✓			

	to specific problems.												
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Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	- Lecture - Discussion - Interaction	- Class Test - Final Exam - Assignment
CLO2	Cognitive (Apply)	- Lecture - Discussion - Interaction	- Class Test - Final Exam - Assignment
CLO3	Cognitive (Create) Affective (Formulate)	- Lecture - Discussion - Interaction	- Class Test - Final Exam - Assignment

Course Code: CSE 1204

Course Title: Object Oriented Programming Sessional

Contact Hours/Week: 3 Hours

Credit Hour: 1.50

Rationale: Object-oriented programming provides a higher-level way for programmers to envision and develop their applications. It provides insights into some of the rudimentary building blocks of software design and development in a module-oriented fashion. As a Computer Science and Engineering major it is essential to learn the concept of object oriented programming in order to provide professional services to the stakeholders including individuals, offices and community.

Course Contents:

Sessional on Object Oriented Programming: Fundamentals of OOP, Classes and Objects, Constructors and Destructors, Inheritance, Polymorphism, Advanced Topics.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understand fundamental concepts of object-oriented programming.	✓											
CLO2	Apply object-oriented programming concepts while writing programs.	✓		✓									
CLO3	Perform individually or as a part of a team while developing object-oriented software.									✓			
CLO4	Design and build innovative software solutions using object-oriented concepts for real life problems.				✓								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO2	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO3	Affective (Respond & Organize)	• Lecture • Tutorial • Discussion • Interaction	• Report • Project • Presentation • Lab Final
CLO4	Cognitive (Apply & Create) Psychomotor (Embody)	• Lecture • Tutorial • Discussion • Interaction	• Report • Project • Presentation • Lab Final

Course Code: EEE 1251

Course Title: Electronic Drives and Circuits

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: The goal of this course is to establish a background on analog circuits, specifically on diode-based rectifiers and their efficiency, BJT, FET and MOSFET-based amplifiers used in discrete circuits and integrated circuits, namely the microelectronic circuits. Different wave shaping techniques of clipping and clamping circuits will also be covered in this course. Furthermore, small signal equivalent circuit models, Small-signal analysis of single-stage amplifiers using BJT will be introduced and analyzed as well. The course also aims to cover the 555 timer, operational amplifiers, gain, input and output stages, as well as the feedback and stability concept in the amplifiers. Basic concepts of oscillators, active filters, voltage regulators, power electronic devices and IC fabrication techniques are covered.

Course Contents:

Introduction to Semiconductors & Diode: P-N Junction Diode, V-I Characteristics, Light-Emitting Diode (LED), Zener Diode, Diode Applications: Half-Wave and Full-Wave Rectifiers – Operation and Efficiency, Ripple factor.

Linear Wave Shaping: Diode Wave Shaping Techniques, Clipping and Clamping Circuits, Voltage Regulation using Zener Diode.

Bipolar Junction Transistor: NPN and PNP Transistors, Amplifying and Switching Actions of Transistor, Transistor Characteristics and Regions of Operation, CB, CE & CC Configurations, Transistor Load Line and Operating Point, BJT Biasing, Small Signal Equivalent Circuit Models, Small-Signal Analysis of Single-Stage Amplifiers, Designing Logic Gate using BJT.

Field Effect Transistor: Principle of Operation of JFET and MOSFET, Depletion and Enhancement Type NMOS and PMOS, MOSFET as switch and amplifier, CMOS Inverter.

Op-amp: Basic OP-amp Characteristics, Gain, Input and Output Impedance, Feedback, Inverting and Non-Inverting Amplifiers, Integrators, Differentiators, Summing Amplifiers, Introduction to Oscillators, Comparator Circuits, Schmitt Trigger, Linear and Non-Linear Applications of Op-amp.

555 Timer: Architecture of 555 Timer, Monostable, Bistable and Astable Multivibrators using 555 Timer.

Filter Circuits: Filter Fundamentals, Different Types of Filters, Passive Filters, Active Filters.

Power Electronic Devices: SCR, TRIAC, DIAC, UJT Characteristics and Applications; Introduction to IC Fabrication Techniques.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understand the functioning of diode and linear wave shaping circuits.	✓	✓										
CLO2	Construct Bipolar Junction Transistor (BJT) circuits.		✓	✓									
CLO3	Analyze Field Effect Transistor (FET) electronic circuits.			✓	✓								
CLO4	Design various electronic circuits using Op-amp and 555 Timer.		✓	✓		✓							
CLO5	Understand filter circuits and power electronic devices.	✓	✓										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	- Lecture - Discussion - Interaction	- Class Test - Final Exam
CLO2	Cognitive (Apply)	- Lecture - Discussion - Interaction	- Class Test - Final Exam
CLO3	Cognitive (Analyze)	- Lecture - Discussion - Interaction	- Class Test - Final Exam
CLO4	Cognitive (Create)	- Lecture - Discussion - Interaction	- Class Test - Final Exam
CLO5	Cognitive (Understand)	- Lecture - Discussion - Interaction	- Class Test - Final Exam

Course Code: EEE 1252

Course Title: Electronic Drives and Circuits Sessional

Contact Hours/Week: 3 Hours

Credit Hour: 1.50

Rationale: Students of the Computer Science & Engineering department need to know analog electronics, especially details about diodes, zener diodes, BJT circuits, MOSFET, op amp, timers, and different timer ICs,

as well as circuit design techniques in order to design efficient analog electronic devices for various applications such as some efficient amplifiers, Oscillators, filters for signal processing units. Besides, students need to learn the design concept of digital logic gates using different types of switching devices.

Course Contents:

Sessional on Electronic Drives and Circuits: Diode, Bipolar Junction Transistor (BJT), Field Effect Transistor (FET), Op-Amps, 555 Timer, Filter Circuit, Logic Gate Design using BJT.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Design and analyze diode circuits.			✓									
CLO2	Determine and analyze response of Bipolar Junction Transistor (BJT) circuits			✓									
CLO3	Determine and analyze response of Field Effect Transistor (FET) electronic circuits				✓								
CLO4	Design and analyze electronic circuits made using Op-amp and 555 Timer.					✓							
CLO5	Design filter and logic gate circuits					✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Create & Analyze)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual • Experiment	• Quiz • Viva • Lab Report • Lab Final • Project
CLO2	Cognitive (Apply & Analyze)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual • Experiment	• Quiz • Viva • Lab Report • Lab Final • Project
CLO3	Cognitive (Apply & Analyze)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual • Experiment	• Quiz • Viva • Lab Report • Lab Final • Project
CLO4	Cognitive	• Lecture	• Quiz

	second order linear differential equations, Maclaurin series and Frobenius series.												
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Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ Level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	- Lecture - Discussion	- Class Test - Final Exam
CLO2	Cognitive (Understand)	- Lecture - Discussion	- Class Test - Final Exam
CLO3	Cognitive (Apply)	- Lecture - Discussion	- Class Test - Final Exam
CLO4	Cognitive (Apply)	- Lecture - Discussion	- Class Test - Final Exam

Course Code: Phy 1213
Course Title: Physics

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: Studying physics strengthens quantitative reasoning and problem solving skills that are valuable in computer science and engineering. This course focuses on basic studies in atomic physics, condensed matter physics, optics, photonics, waves and oscillations and allied areas. The physics and engineering physics are great preparation for almost any career, because they teach students how to analyze complex problems and they give students a strong quantitative background that can be applied in any technical field.

Course Contents:

Structure of Matter: Structure of Matter. Different Types of Bonds in Solids: Metallic, Van Der Waals', Covalent and Ionic Bond. Packing in Solids: Inter Atomic Distances and Forces of Equilibrium, X-Ray Diffraction, Bragg's Law, Distinction among Insulator, Semiconductor and Conductor.

Atomic Physics: Atom Models: Thomson Atom Model, Rutherford Atom Model, Rutherford Scattering Formula, Electron Orbits, Bohr Atom Model, Energy Levels and Spectra, Particle Properties of Waves: Photoelectric Effect, Einstein's Photoelectric Equation, Laws of Photoelectric Emission, Photovoltaic Cells, Compton Effect, Wave Properties of Particle: De Broglie Waves, Group Velocity, Phase Velocity.

Waves and Oscillations: Oscillations: Simple Harmonic Motion, Composition of Simple Harmonic Motions and Lissajous' Figures, Damped and Forced Oscillations, Resonance. Waves: Traveling and Standing waves, Energy Calculation of Traveling and Standing Waves, Intensity of Waves, Beats, Doppler Effect.

Physical Optics: Theories of Light: Wave theory: Huygen's Wave Theory, Huygen's Principle and Construction, Superposition of Light Waves, Electromagnetic Theory, Particle Theory: Newton's Corpuscular Theory, Quantum Theory of Light.

Interference: Introduction, Conditions of Interference, Young's Double Slit Experiment, Fresnel's Bi-prism, Thin Film Interference, Interference Due to Multiple Reflection, Newton's Ring.

Diffraction: Fresnel and Fraunhofer Diffraction, Diffraction by Single and Double Slit, Diffraction Gratings.

Polarization: Introduction, Methods of Producing Polarized Light, Polarization by Reflection and Refraction, Polarization by Double Refraction, Construction of Nicol Prism. Production and Analysis of Polarized Light, Optical Activity, Optics of Crystals, Polarimeter.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

[illegible]

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam
CLO2	Cognitive (Understand)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam ● Assignment
CLO3	Cognitive (Apply)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam ● Assignment

Course Code: Phy 1214
Course Title: Physics Sessional

Contact Hours/Week: 3/2 Hours
Credit Hour: 0.75

Rationale: This sessional course is designed to demonstrate measuring instruments that are typical in physics laboratory for engineering applications. The program is also designed to develop strong mathematical and analytical skills, good laboratory skills, and effective written and oral communication skills. This course also helps to strengthen quantitative reasoning and problem-solving skills that are valuable in computer science and engineering.

Course Contents:

Sessional on Physics: Introduction to Laboratory Works and Experiments, Waves and Oscillations, Atomic and Modern Physics, Physical Optics.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

[illegible]

	regarding mechanics, light and electricity.												
CLO2	Understand the working of different instruments and instrumentation used in the experiments.	✓											
CLO3	Apply these instruments to measure the various physical parameters relating waves and oscillations, atomic and modern physics, physical optics.		✓										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	● Lecture ● Tutorial ● Discussion ● Interaction ● Lab Manual	● Quiz ● Viva ● Lab Report ● Lab Final
CLO2	Cognitive (Understand)	● Lecture ● Tutorial ● Discussion ● Interaction ● Lab Manual	● Quiz ● Viva ● Lab Report ● Lab Final
CLO3	Cognitive (Apply)	● Lecture ● Tutorial ● Discussion ● Interaction ● Lab Manual	● Quiz ● Viva ● Lab Report ● Lab Final

2nd YEAR 1st SEMESTER

Course Code: CSE 2100

Course Title: Software Development Project I

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: This course is designed to develop an application program with an emphasis on software development strategies. Integrated development tools, software development kits, and software subsystems are to be employed to develop one or more Programs / Projects on some practical problems. Learning activities in this course mainly include tasks related to developing the knowledge and skills necessary to write effective computer programs for information system applications.

Course Contents:

Students will develop one or more Programs/Projects on some practical problems with sound Software Engineering practices as assigned by the teacher.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding the requirements to develop a software.					✓							
CLO2	Selecting project ideas and designing prototype.							✓					
CLO3	Implementing the idea in project development.									✓		✓	
CLO4	Presenting the project.										✓		

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction • Audio/Video	• Viva • Project
CLO2	Cognitive (Apply)	• Discussion • Interaction	• Viva • Project
CLO3	Cognitive (Apply)	• Discussion • Interaction	• Viva • Project
CLO4	Psychomotor (Embody)	• Discussion • Interaction	• Viva • Project • Presentation

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Course Contents:

Set: Using Set Notation with Quantifiers, Truth Sets and Quantifiers, Generalized Unions and Intersections, Computer Representation of set.

Sequences and Summations: Sequences, recurrence relations and Summations.

Cardinality of Sets: Countable Sets, Uncountable Set, Uncomputable Functions.

Number Theory: Divisibility and Modular Arithmetic, Primes and Greatest Common Divisors, Solving Congruences, Applications of Congruences.

Induction and Recursion: Mathematical Induction, Strong Induction and Well-Ordering, Recursive Definitions and Structural Induction.

Counting: The Pigeonhole Principle, Binomial Coefficients and Identities, Generalized Permutations and Combinations, Generating Permutations and Combinations. Applications of Recurrence Relations, Generating Functions, Inclusion–Exclusion, Applications of Inclusion–Exclusion.

Relation: Property of relation, Representing Relations, Closures of Relations, Equivalence Relations, Partial Orderings.

Graphs: Graphs and Graph Models Graph Terminology and Special Types of Graphs, Representing Graphs, Graph Traversing, Graph Isomorphism, Connectivity, Euler and Hamilton Paths, Planar Graphs, Graph Coloring.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

[illegible]

CLO2	Know basic properties of set, function & relations.	✓											
CLO3	Analyze different problems based on number theory, recursion and counting method.		✓										
CLO4	Know essential concepts in graph theory and analyze related algorithms.	✓	✓										
CLO5	Apply knowledge about discrete mathematics in problem solving.			✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand & Analyze)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Lab Final • Project
CLO2	Cognitive (Remember)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Lab Final • Project
CLO3	Cognitive (Analyze)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Lab Final • Project
CLO4	Cognitive (Remember and Analyze)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Lab Final • Project
CLO5	Cognitive (Apply)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Lab Final • Project

Course Code: CSE 2103
Course Title: Digital Logic Design

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: All real-world data are analog and these data are communicated from one place to another through digital devices, whereas almost all kinds of digital devices like computer, calculator, digital machine, mobile phone etc. use digital data. For this purpose and for designing various digital devices, digital logic design course is very important to implement the desired devices.

Course Contents:

Information and Digital Systems: Introduction to Digital Systems, Number Systems, Weighted and Non-Weighted Codes, Error Detection Code, Binary Addition and Subtraction, 2's Complement Methods.

Boolean Algebra and Combinational Logic Circuits: Digital Logic, Boolean Algebra, Boolean Function, Canonical Forms, Karnaugh Maps, Minimization of Boolean Functions, Logic Gates and their Truth Tables, Design Methodologies, Combinational Logic Circuit Design, Arithmetic and Data Handling Logic Circuits. Decoders, Encoders, Multiplexer, Demultiplexer.

Flip Flop and Sequential Logic Circuits: Transistor Latch, NAND Gate Latch, NOR Gate Latch, D Latch. Clock Signals and Clocked FFs: Clocked SR, JK and D Flip-Flops, Master/Slave JK FF, Timing Diagram of Different FFs, Edge-Triggered and Level-Triggered Timing Diagrams, Counters, Registers, Memory Devices and their Applications.

Hardware Description Language (HDL): Synthesis of digital circuits using Hardware Description Language (HDL). Reference Language: Verilog.

Technology Parameters: Fan In, Fan Out, Propagation Delay, Power Dissipation and Noise Immunity.

Others: Diode Logic Gates, Transistor Gates, MOS Gates, Logic Families: TTL and CMOS Logic with Operation Details.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding various number systems and their conversions, representation of numbers in digital devices.		✓										
CLO2	Developing a combinational digital logic circuit and applying it to solve real life problems.		✓										
CLO3	Analyzing and designing sequential logic circuits.			✓									
CLO4	Applying HDL to design combinational and sequential logic circuits.			✓									
CLO5	Designing and implementing logic circuits using TTL & CMOS families.				✓								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Apply & Create)	• Lecture	• Class Test

		• Discussion • Interaction	• Final Exam
CLO3	Cognitive (Analyze)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment • Presentation
CLO4	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Project
CLO5	Cognitive (Apply & Create)	• Lecture • Discussion • Interaction • Audio/Video	• Class Test • Final Exam • Project

Course Code: CSE 2104

Course Title: Digital Logic Design Sessional

Contact Hours/Week: 3 Hours

Credit Hour: 1.50

Rationale: All real-world data are analog and these data are communicated from one place to another through digital devices, whereas almost all kinds of digital devices like computer, calculator, digital machine, mobile phone etc. use digital data. For this purpose and for designing various digital devices, digital techniques course is very important to implement the desired devices.

Course Contents:

Sessional on Digital Logic Design: Designing Various Logic Combinational circuits, Designing Various Sequential Circuits, Hardware Description Language (HDL), Designing and Implementing Various TTL & CMOS Logic Devices. Reference HDL: Verilog.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Designing and implementing various combinational logic circuits for solving the practical problems.			✓				✓					
CLO2	Designing and implementing various sequential logic circuits for solving the practical problems.			✓				✓					
CLO3	Applying HDL to design combinational and sequential logic circuits.			✓				✓					
CLO4	Designing and implementing various gates by using TTL & CMOS logic.			✓				✓					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Apply & Create)	● Lecture ● Discussion ● Interaction ● Lab Manual	● Quiz ● Viva ● Lab Report ● Lab Final ● Project
CLO2	Cognitive (Apply & Create)	● Discussion ● Interaction ● Lab Manual	● Quiz ● Viva ● Lab Report ● Lab Final ● Project
CLO3	Cognitive (Apply)	● Discussion ● Interaction ● Lab Manual	● Quiz ● Viva ● Lab Report ● Lab Final ● Project
CLO4	Cognitive (Apply & Create)	● Lecture ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Lab Report ● Lab Final ● Project

Course Code: EEE 2151**Course Title: Electrical Drives and Instrumentation****Contact Hours/Week: 3 Hours****Credit Hour: 3.00**

Rationale: Students in the Department of Computer Science & Engineering need to study Electrical Machines and Instrumentations course to be familiar with the intuitive concepts of the fundamental electromagnetic laws, measuring instruments, transducers etc. that control the operation of electrical machinery such as transformers, generators, and motors etc. Furthermore, deep insight relating to construction, modeling, analysis, detailed working and modern day applications of mentioned electrical machines and instruments will be carried out so that the students will be able to cope with different industrial machineries.

Course Contents:

DC Machines: Operation and Performance Characteristics of Generators and Motors, Starting, Speed Control and Braking of Motors, Different Application of DC Motors.

AC Machines: Transformer: Principle of Operation of Transformer, Single Phase Induction Motors and its Methods of Starting, Introduction to alternator.

Special Machines: Stepper Motor, Servomotor, Brush less Motors.

Measuring Instruments: Electromechanical and Electronic Meters and their Uses, Extension of Instrument Range.

Instrumentation Amplifiers: Differential, Logarithmic and Chopper Amplifiers, Frequency and Voltage Measurements using Digital Techniques, Recorders and Display Devices, Spectrum Analyzers and Logic Analyzers.

Transducers: Terminology, Types, Principles and Application of Photovoltaic, Piezoelectric, Thermoelectric, Position and Displacement Transducers, Potentiometer, Linear Variable Differential Transformers (LVDT), Pressure Transducer, Temperature Transducer, Optical Transducer, Ultrasonic Transducer, Humidity Transducer, Hall Effect Transducer, Speed Transducer, Noise Reduction in Instrumentation.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	To understand the intuitive concepts of the fundamental electromagnetic laws.	✓	✓										
CLO2	To understand the construction, operation and analysis of DC Generators and Motors.	✓	✓										
CLO3	To understand the construction, operation and analysis of AC Generators and Motors and Special Machines.	✓	✓										
CLO4	To learn techniques involved in analyzing measurement data and the errors associated with the measurement system used.		✓		✓								
CLO5	To measure various physical quantities using instrumentation amplifiers and transducers.				✓								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO2	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO3	Cognitive (Analyze)	• Lecture • Tutorial • Discussion • Interaction	• Class Test • Final Exam • Assignment

CLO4	Cognitive (Analyze)	• Lecture • Tutorial • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO5	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction	• Class Test • Final Exam • Assignment

Course Code: EEE 2152

Contact Hours/Week: 3/2 Hours

Course Title: Electrical Drives and Instrumentation Sessional

Credit Hour: 0.75

Rationale: The basic concepts, mathematical models, operating principles and specially the practical knowledge to operate and test the mostly used electrical machines, transformers, and measuring instruments in real life engineering applications, are highly recommended for the students of Computer Science & Engineering. Moreover, the technical skills to handle any troubleshoot when operating these machines and instruments are greatly needed. Taking all these into consideration, the course has been designed, so that the students will be able to understand and apply the technical skills in real life engineering applications.

Course Contents:

Sessional on Electrical Drives and Instrumentation: Transformer, DC Machine, AC Machine, Measuring Instruments and Transducers.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Apply the knowledge to test the transformer in the field.	✓	✓										
CLO2	Apply the knowledge to operate the dc and ac machines in practical applications.	✓			✓								
CLO3	Apply the knowledge to use the measuring instruments and transducers in practical applications.	✓	✓										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Lab Final • Project

CLO2	Understand vector differentiation and integration.	✓	✓										
CLO3	Determine the normal form, adjoint, inverse, eigen values and eigen vectors of the matrices, determine the solution of system of linear equation.	✓	✓										
CLO4	Understand about vector space, determine basis and dimension of vector space, and linear mapping.	✓	✓										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO3	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO4	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam

Course Code: Hum 2113

Course Title: Economics, Government and Sociology

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: This course is designed to give undergraduate level fundamental knowledge of economic principles and methods which will enable them to understand the socio-economic environment. This course emphasizes the application of basic Microeconomic and Macroeconomic concepts to current economic events. A combination of theory and practice will be stressed. To the greatest extent possible, the course will be tailored towards exuding engineering economics concepts. In addition, the student will be able to compare and contrast a sociological perspective with other scientific perspectives (society; social structure; culture; religion; social institutions, e.g., family, economy; social change; social class; status; race; ethnicity; gender; social conflict; deviance; cyber crime etc.).

Course Contents:

Economics

Introduction of Economics: Definition of Economics, Goals of Economics, Scope of Economics, Economic Problems, Why Study Economics, Scarcity & Efficiency, Economic & Technical Efficiency.

Demand, Supply and Their Elasticity: Definition of Demand, Law of Demand and Its Limitations, Demand Curve, Market Demand, Determinants of Demand, Extension & Contraction of Demand, Increase and Decrease of Demand, Elasticity of Demand and its Types, Elastic & Inelastic Demand, CPE, IE, Determinants of Elasticity of Demand, Total Revenue Test, Elasticity & Revenue, Point Elasticity, Elasticity in a Linear Demand Curve, Supply and Stock, Law of Supply and its Limitations, Supply Curve, Shifts in Supply Curve, Determinants of Supply, Market Supply, Elasticity of Supply. Consumer Surplus & Producer Surplus, Equilibrium, Shortage and Surplus.

Theory of Production: Factors of Production, Producer's Equilibrium - Isoquant Curve, Production Possibility Curve - Equilibrium of a Firm, Internal & External Economies & Diseconomies, Laws of Return, Input Output Analysis.

Theory of Cost: Cost of Production, Cost Function, Total Cost-Fixed & Variable, AC, MC, Why Short Run Average Cost Curve is U-Shaped?, Envelope Curve or Long Run Cost Curve, Explicit & Implicit Cost, Engineering Cost Curves, Math.

Economic Development and Economic Planning: Economic Growth, Economic Growth Rate and When it will be Double - Calculation, Economic Development, Difference Between Economic Growth and Development, Indicators of Economic Development, HDI, Barriers to Economic Development in Developing Countries, Sustainable Development & its Goals, Economic Planning, Essential Features of Economic Planning, Objectives of Economic Planning, Stages of Planning, Essentials of Planning, Planning in Bangladesh.

Concept of National Income: Define National Income. Circular Flow of NI in Two Sectors. GDP, GNP, NNP, NDP, PI, DPI, Saving, Investment, Methods of Calculating NI, Methods used in Bangladesh, Problems of Calculating NI in Bangladesh, Importance of Calculating NI, Math.

Economic Policies: Fiscal Policy, Monetary Policy, Trade Policy and their Applicability in Bangladesh, Industry & Population of Bangladesh.

Government

Institutions: State - Forms & Functions of State, Government- Forms & Functions of Government, Local & Central Government, Judiciary & Legislative System, Political, Economic & Recreational Institutions and their Roles in the Society, Formal Organization & Bureaucracy, Public Opinion.

Sociology

Introduction: Define Society, Evaluation of Society, Distinguish between Community and Society, Culture, Define Sociology, Importance of Sociology, Scope of Sociology.

Social System: Family, Marriage, Role of Engineers in the Society, Effects of Change in Technology on Society, Ethical Perspectives of Technology.

Culture: Elements of Culture, Cultural Systems and Subsystems, Cultural Lag, Relationship between Culture and Civilization.

Social Problems: Unemployed Delinquency and Crime, Social Crime, Cyber Crime, Deviant Behavior, Globalization and Human Rights.

Others: Geographical Location of Bangladesh, Urban Community, Urban Ecology, Rural & Urban Power Structure.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
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CLO1	Understand major principles of economic analysis for decision making among alternative courses of action in engineering.	✓										✓	
CLO2	Apply economic principles to prices and quantities in competitive supply and demand for goods market and for money.											✓	✓
CLO3	Solve economic problems involving comparison and selection of alternatives by using analytical techniques including benefit-cost ratio and breakeven analysis.							✓				✓	
CLO4	Describe processes of socialization, and how socialization operates in different societies and cultures and methods of social control, including political & legal systems.						✓			✓			
CLO5	Describe how the tools of analysis and methods of sociology are applicable to work and involvement in their community and be able to explain the concept of deviance & cyber crime.						✓		✓				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Presentation
CLO3	Cognitive (Apply)	• Lecture • Discussion	• Class Test • Final Exam

		● Interaction	● Assignment
CLO4	Cognitive (Understand)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam ● Assignment
CLO5	Cognitive (Understand)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam ● Assignment

2nd YEAR 2nd SEMESTER

Course Code: CSE 2200

Course Title: Technical Writing and Presentation Sessional

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: Technical writing is a type of writing in which the author addresses a specific topic that requires direction, instruction, or explanation. Technical writing skills are necessary for preparing quality research papers, books, theses, reports, reviews, etc. Moreover, technical writing skills are also important for preparing proper documentation of a product or discovery in the domain of Computer Science & Engineering. On the other hand, the ability to provide successful and compelling presentations to a variety of audiences is known as presenting skills. Therefore, keeping the mentioned necessities in mind, this course will cover both technical writing and presentation.

Course Contents:

Sessional on Technical Writing and Presentation:

Introduction: Issues of Technical Writing and Effective Oral Presentation in Computer Science and Engineering; Criteria for Good Technical Writing and Presentation; An Overview of Research Methodologies: Quantitative and Qualitative Research.

Writing Issues: Writing Styles of Definitions, Propositions, Theorems, Proofs, etc.; Preparation of Reports, Scientific Articles, and Research Papers.

Thesis, Books and Others: Abstract, Preface, Contents, Bibliography and Index; Writing of Book Reviews and Referee Reports; Curriculum Vitae, Resume Writing, etc.

Writing and Presentation Tools: LATEX; Diagram Drawing Software; Presentation Tools.

Writing Ethics: Plagiarism Issues; Writing and Presentation Ethics; Ethical Technical Communication.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Communicate effectively on complex activities with the research community and with society at large such as write effective reports, prepare technical papers, design documentation, and make effective presentations.										✓		
CLO2	Prepare documentation, technical papers, reports, and reviews in multi-disciplinary settings both as an individual member or as a team player/leader.									✓			
CLO3	Understand the curse of plagiarism, the								✓				

	ways to overcome plagiarism, and the process of maintaining ethics in the field of technical writing and presentations.												
CLO4	Learn to write documentation, presentations, research articles, and reviews by utilizing modern tools i.e., LATEX, diagram drawing software, presentation tools, plagiarism checkers, etc.					✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Lab Final • Project
CLO2	Affective (Organize)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Lab Final • Project
CLO3	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Lab Final • Project
CLO4	Psychomotor (Embody)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Lab Final • Project • Presentation

Course Code: CSE 2201

Course Title: Algorithm Analysis and Design

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: Algorithm design and analysis are essential in the daily work of a skilled programmer because they provide the theoretical backbone of computer science. This course aims to give a solid foundation in the design and analysis of the major classes of algorithms. Students will be able to create their own versions of a specific computational task and compare their performance at the end of the course.

Course Contents:

Methods of Complexity Analysis: Substitution method, Iteration Method, Master Method, Amortized Analysis.

Linear Time Sorting: Radix Sort, Bucket Sort, Counting Sort, Proof of Correctness of Sorting algorithms, Lower bound on sorting

Divide and Conquer approach: Minimum and Maximum Elements Finding, Median Finding, Karasuba multiplication: Integer Multiplication, Strassen's Matrix Multiplication, Quick Sort, Divide and Conquer Based Sorting Algorithms Review.

Greedy Algorithms: Greedy Algorithm Properties, Job Sequencing Problem, Huffman Codes, Fractional Knapsack Problem, Minimum Spanning Tree: Kruskal and Prim's Algorithm, Single Source Shortest Path: Dijkstra's Algorithm, Proof of correctness

Dynamic Programming: Dynamic Algorithm Properties, Longest Common Subsequence (LCS), 0-1 Knapsack, Knapsack with duplicate items, Max Sum without Adjacent, Matrix Chain Multiplication (MCM), Single Source Shortest Path: Bellman-Ford Algorithm, All pair of shortest path: Floyd Warshall Algorithm, Bitmasking with DP: Traveling Salesperson Problem.

Backtracking: Backtracking, State Space Tree, N-Queen Problem, M-Coloring, Rat in a Maze, Solving Sudoku / Puzzle.

Branch and Bounds: Branch and Bounds in State Space Tree, LC Branch and Bound, FIFO Branch and Bound.

Flow Networks: Maximum Flow, Minimum Cut problem, Ford-Fulkerson Algorithm

P and NP Problems: Introduction to Optimization and Decision Problems, Tractable and Intractable Problems, Deterministic and Nondeterministic Algorithms, P and NP Set of problems, NP-hardness and NP-completeness, Proof of NP-completeness for algorithms, Cook's Theorem.

Reducibility between Problems and Approximation Algorithms: Discussion of different NP-complete problems like Satisfiability, Clique, Vertex Cover, Independent Set, Hamiltonian Cycle, TSP and Knapsack. Reduction between problems, Approximation algorithm design for sample problems and corresponding performance bound analysis.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

[illegible]

CLO4	Analyze the suitability of using particular data structures for computational problems and apply greedy and dynamic algorithm techniques to different real world problems.		✓	✓	✓								
CLO5	Apply backtracking, graph algorithms or randomized algorithmic design paradigms to solve computational problems by identifying key characteristics of a problem.	✓	✓	✓									
CLO6	Synthesize efficient algorithms in common engineering design situations.	✓		✓	✓								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Evaluate)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam
CLO2	Cognitive (Apply & Analyze)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam ● Presentation
CLO3	Cognitive (Apply & Analyze)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam ● Assignment
CLO4	Cognitive (Analyze)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam ● Assignment
CLO5	Cognitive (Apply)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam ● Assignment
CLO6	Cognitive (Create)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam ● Assignment

Course Code: CSE 2202

Course Name: Algorithm Analysis and Design Sessional

Contact Hours/Week: 3 Hours

Credit Hour: 1.50

Rationale: Algorithm design and analysis are essential in the daily work of a skilled programmer because they provide the theoretical backbone of computer science. This course aims to give a solid foundation in the design and analysis of the major classes of algorithms. Students will be able to create their own versions of a specific computational task and compare their performance at the end of the course.

Course Contents:

Sessional on Algorithm Analysis and Design: Methods of Complexity Analysis, Linear Time Sorting, Divide and Conquer Approach, Greedy Algorithms, Dynamic Programming, Backtracking, Branch and Bounds, Flow Networks, P and NP Problems, Reducibility between Problems and Approximation Algorithms.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Evaluate the performance of computer algorithms and develop correctness proofs for algorithms.		✓		✓	✓							
CLO2	Implement and analyze linear time sorting algorithms.		✓	✓									
CLO3	Implement and analyze Divide-conquer algorithms from the perspective of several real world problems.	✓	✓										
CLO4	Analyze the suitability of using particular data structures for computational problems and apply greedy and dynamic algorithm techniques to different real world problems.		✓	✓	✓								
CLO5	Apply backtracking, graph algorithms or randomized algorithmic design paradigms to solve computational problems by identifying key characteristics of a problem.	✓	✓	✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
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CLO1	Cognitive (Evaluate)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Lab Report ● Lab Performance ● Lab Final ● Project
CLO2	Cognitive (Apply & Analyze)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Lab Report ● Lab Performance ● Lab Final ● Project
CLO3	Cognitive (Apply & Analyze)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Lab Report ● Lab Performance ● Lab Final ● Project
CLO4	Cognitive (Analyze)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Lab Report ● Lab Performance ● Lab Final ● Project
CLO5	Cognitive (Apply)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Lab Report ● Lab Performance ● Lab Final ● Project

Course Code: CSE 2203
Course Title: Numerical Methods

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: Numerical methods are techniques for formulating mathematical problems so that they can be solved using arithmetic operations. This course covers the more common and powerful numerical methods and techniques appropriate for bachelor's level engineering technology students. Root finding, optimization, curve fitting, solution of simultaneous equation, and numerical differentiation/integration are common topics. A course of this type should also expose students to commercial software that can demonstrate techniques and solve problems that cannot be solved by hand.

Course Contents:

Modeling, Computers and Error Analysis: Mathematical Modeling and Engineering Problem Solving, Programming and Software, Approximations and Round-Off Errors, Truncation Errors and the Taylor Series.

Roots of Equations: Bracketing Methods, Open Methods, Roots of Polynomials.

Linear Algebraic Equations: Gauss Elimination, LU Decomposition and Matrix Inversion, Gauss-Seidel.

Optimization: One-Dimensional Unconstrained Optimization.

Curve Fitting: Least-Square Regression.

Interpolation: Interpolation with one and two Independent Variables, Formation of Different Difference Table, Newton's Forward and Backward Difference, Lagrange's Interpolation.

Numerical Differentiation and Integration: Newton-Cotes Integration Formulas, Integration of Equations, Numerical Differentiation

Ordinary Differential Equations: Runge-Kutta Methods, Boundary-Value and Eigenvalue Problems, Numerical Solution of Partial Differential Equations

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding various types of error, root finding methods of different degree of equations in numerical analysis.		✓										
CLO2	Analyzing different linear algebraic equations and optimization problems according to problem domain.		✓										
CLO3	Analyzing and Implementing various types of interpolation method and Least Square curve fitting technique.			✓									
CLO4	Evaluating numerical differentiation and Integration techniques, ordinary differential equations with appropriate examples.		✓										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO2	Cognitive (Analyze)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO3	Cognitive (Analyze and Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO4	Cognitive (Evaluate)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment

Course Code: CSE 2204
Course Title: Numerical Methods Sessional

Contact Hours/Week: 3/2 Hours
Credit Hour: 0.75

Rationale: Numerical methods are techniques for formulating mathematical problems so that they can be solved using arithmetic operations. This course covers the more common and powerful numerical methods and techniques appropriate for bachelor's level engineering technology students. Root finding, optimization, curve fitting, solution of simultaneous equation, and numerical differentiation/integration are common topics. A course of this type should also expose students to commercial software that can demonstrate techniques and solve problems that cannot be solved by hand.

Course Contents:

Seasonal on Numerical Methods: Modeling, Computers and Error Analysis, Roots of Equations, Linear Algebraic Equations, Optimization, Curve Fitting, Interpolation, Numerical Differentiation and Integration, Ordinary Differential Equations.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Implementing various types of error, root finding methods of different degrees of equations in numerical analysis.		✓										
CLO2	Analyzing and implementing various types of interpolation method and Least Square curve fitting technique.			✓									
CLO3	Evaluating numerical differentiation and Integration techniques, ordinary differential equations with appropriate examples.			✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Apply)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Lab Final • Project
CLO2	Cognitive (Analyze & Create)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Lab Final • Project
CLO3	Cognitive (Evaluate)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Lab Final • Project

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Course Code: CSE 2205

Course Title: Microprocessors, Microcontrollers and Assembly Language

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: The most common methods of integrating intelligence into automated devices have been microprocessors and assembly language. As a result, it is critical to understand how they work and how they can be used as building blocks for automated systems and control applications. This course investigates the inner workings of a microprocessor from a programmer's standpoint, and several laboratory exercises will be based on a microprocessor using the assembly language.

Course Contents:

Microprocessor: Introduction to Different Types of Microprocessors and its Applications, Organization of Intel x86 Architecture, Overview of Different Generations of Intel Processors, Component of Microcomputer System, Interrupt Structures, I/O Interfacing, DMA, Co-processors, Memory Module, Overview of Advanced Intel Instructions: AVX, MMX, SSE & SIMD.

Microprocessor Assembly: Introduction to x86 and x86-64 Assembly, Assembly Program Structure and Its Components, Basic Instructions, Input/Output Instructions, Flag Register, Flow Control Instructions, Logic, Shift and Rotate Instruction, Arithmetic Instructions, Arrays and Related Addressing Modes, Basic Stack Operations, Procedures Declaration, Calling a Procedure, String Instructions, Macros, Floating-Point Instructions.

Microcontroller: Microcontroller History and Features, Microcontroller Architecture, General Purpose Registers in Microcontroller, Microcontroller Data Memory, Microcontroller Status Register, Microcontroller Data Format and Directives.

Microcontroller Assembly: Microcontroller Assembly Programming, Program Counter and Program ROM Space in Microcontroller, Branch Instructions and Looping, Call Instructions and Stack, Time Delay, Arithmetic Instructions, Logic and Compare Instructions, Rotate and Shift Instructions.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Explain the internal architecture of different microprocessors and their addressing modes.	✓											
CLO2	Apply microprocessor assembly code to use common instructions		✓										
CLO3	Apply microprocessor assembly code to use advanced instructions.					✓							

CLO4	Understanding microcontroller architecture and its internals.	✓											
CLO5	Apply microcontroller assembly code to use basic instructions, branch, call and time delay instructions.					✓							
CLO6	Apply microcontroller assembly code to use arithmetic and logic instructions.					✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction • Audio/Video	• Class Test • Final Exam
CLO2	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO3	Cognitive (Apply)	• Lecture • Discussion • Interaction • Audio/Video	• Class Test • Final Exam
CLO4	Cognitive (Understand)	• Lecture • Discussion • Interaction • Audio/Video	• Class Test • Final Exam
CLO5	Cognitive (Apply)	• Lecture • Discussion • Interaction • Audio/Video	• Class Test • Final Exam
CLO6	Cognitive (Apply)	• Lecture • Discussion • Interaction • Audio/Video	• Class Test • Final Exam

Course Code: CSE 2206

Course Title: Microprocessors, Microcontrollers and Assembly Language Sessional

Contact Hours/Week: 3 Hours

Credit Hour: 1.50

Rationale: The most common methods of integrating intelligence into automated devices have been microprocessors and assembly language. As a result, it is critical to understand how they work and how they can be used as building blocks for automated systems and control applications. This course investigates the inner workings of a microprocessor from a programmer's standpoint, and several laboratory exercises will be based on a microprocessor using the assembly language.

Course Contents:

Sessional on Microprocessors, Microcontrollers and Assembly Language: Environment Setup, Microprocessor Assembly: Registers and Program Structure, Flow Control, Looping and Branching, Logic and Arithmetic Operation, Stack Operation, Arrays and Data Structure, Addressing Modes, String Manipulation, Macros, Microcontroller Assembly: Branch and Looping Instructions, Call Instructions and Stack, Time Delay, Arithmetic Instructions, Logic and Compare Instructions, Rotate and Shift Instructions.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Design and analyze microprocessor assembly code to use different arithmetic & logical operations.		✓										
CLO2	Design microprocessor assembly code to use the branching structures, looping structures and analyze the effects on flags.		✓			✓							
CLO3	Design microprocessor assembly code to use floating point instructions.					✓							
CLO4	Design and analyze microprocessor assembly code to use stacks, procedures, string operations, and macros.		✓			✓							
CLO5	Design and analyze microcontroller assembly code to use basic instructions, branch, call and time delay instructions.					✓							
CLO6	Design and analyze microcontroller assembly code to use arithmetic and logic instructions.					✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Create & Analyze)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Lab Assessment • Quiz • Viva • Lab Report • Lab Final

			• Project
CLO2	Cognitive (Create & Analyze)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Lab Assessment • Quiz • Viva • Lab Report • Lab Final • Project
CLO3	Cognitive (Create)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Lab Assessment • Quiz • Viva • Lab Report • Lab Final • Project
CLO4	Cognitive (Create & Analyze)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Lab Assessment • Quiz • Viva • Lab Report • Lab Final • Project
CLO5	Cognitive (Create & Analyze)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Lab Assessment • Quiz • Viva • Lab Report • Lab Final • Project
CLO6	Cognitive (Create & Analyze)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Lab Assessment • Quiz • Viva • Lab Report • Lab Final • Project

Course Code: Math 2213

Course Title: Complex Variable, Partial Differential Equation and Harmonic Analysis

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: Complex analysis is the branch of mathematical analysis that investigates functions of complex numbers. It is very helpful in theoretical computer science, covering property testing, communication complexity and PAC learning. Differential equation may be used in computer science to model complex interaction or nonlinear phenomena. Students used PDE in Computer Graphics and Geometric Processing, game apps development, robotic industrialization. Harmonic analysis is a branch of mathematics concerned with the representation of functions or signals as the superposition of basic waves. Harmonic Analysis can be very useful in discrete mathematics and theoretical computer science, signal processing, machine learning and data analysis. These techniques deliver the answers desired by engineers working with big data sets or searching for novel methods to connect experiments with theory. In particular, the Harmonic Analysis provides tools for filtering signals based on amplitude-frequency characteristics.

Course Contents:

Complex Variable: Complex Number Systems, General Functions of a Complex Variable, Limits and Continuity of a Function of Complex Variable and Related Theorems, Complex Differentiation and the Cauchy-Riemann Equations, Infinite Series, Convergence, Line Integral, Cauchy Integral Theorem, Cauchy

Integral Formula, Liouville's Theorem, Taylor's and Laurent's Theorems, Singular Points, Residue, Cauchy's Residue Theorem, Contour Integration.

Partial Differential Equation: Partial Differential Equations, Solution of First Order Partial Differential Equation by Lagrange and Charpit Methods, Solution of Laplace Equation and Wave Equation.

Harmonic Analysis and Laplace Transform: Fourier Series and Fourier Transformations and its Applications to Solve Boundary Value Problems. Laplace Transforms, Inverse Laplace Transforms, Solution of Differential Equation by Laplace Transforms.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understand the basic concept of complex number system, function of complex variables, and analyze the limit, continuity and differentiability of complex variable functions.	✓											
CLO2	Evaluate complex integration applying Cauchy's theorems and residues theorems.	✓	✓										
CLO3	Understand Taylor, power and Laurent series, and classify singularities and poles.	✓	✓										
CLO4	Understand PDE and solve 1 st order linear & non-linear PDE and also solution of Laplace and Wave Equation with application.		✓										
CLO5	Understand the behavior of periodic function and represent it using Fourier series and also understand the Fourier Transform to solve boundary value problems.	✓	✓										
CLO6	Understand the fundamentals of the Laplace transform and apply Laplace transforms to solve initial and boundary value problems.	✓											

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam

	flow analysis for the industry.												
CLO3	Solve real life problems relating to plant layout, inventory control, performance management, training and motivation.									✓		✓	
CLO4	Calculate various cost accounting metrics.											✓	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Remember & Understand)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam
CLO2	Cognitive (Apply)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam ● Assignment ● Presentation
CLO3	Cognitive (Apply)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam ● Assignment
CLO4	Cognitive (Apply)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam ● Assignment ● Presentation

3rd YEAR 1st SEMESTER

Course Code: CSE 3100

Course Title: Web Based Application Project

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: A web-based application is any program that is accessible via a network connection rather than being stored in the memory of a device. Web-based apps are frequently run within a web browser. However, web-based programs can also be client-based, in which a tiny portion of the software is downloaded to a user's desktop but processing is done on an external server through the internet.

Course Contents:

Students will work in groups or individually to develop web-based applications and a website by incorporating client-side and server-side scripting and interfacing the web applications to a database.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Apply modern tools to design and create a useful web application.					✓							
CLO2	Generate web application ideas that can contribute greatly for the betterment of the society and environment.						✓	✓					
CLO3	Organize team members to obtain the best outcomes from them.									✓			
CLO4	Learn about project and finance management.											✓	✓

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Apply)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Project • Report • Viva
CLO2	Affective (Internalize)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Project • Report • Viva
CLO3	Cognitive (Organize)	• Tutorial • Discussion	• Cont. Asses. • Project

CLO2	Apply the major components of the database models to design Relational databases using the E-R model and normalization approach.					✓							
CLO3	Demonstrate the basics of SQL and construct queries using SQL and introduction to relational database theory and be able to write relational algebra expressions for queries.		✓										
CLO4	Implement the principles and concepts of information integrity, security and data recovery.							✓					
CLO5	Understand advanced database management system concepts.		✓										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Class Test • Final Exam • Assignment
CLO2	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Class Test • Final Exam • Assignment
CLO3	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Class Test • Final Exam • Assignment
CLO4	Cognitive (Create)	• Lecture • Tutorial • Discussion • Interaction	• Class Test • Final Exam • Assignment

		• Audio/Video	
CLO5	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Class Test • Final Exam • Assignment

Course Code: CSE 3102

Course Title: Database Systems Sessional

Contact Hours/Week: 3 Hours

Credit Hour: 1.50

Rationale: With the lessons of database systems, students will be able to design, test, and implement analytical, transactional database systems according to business requirements. This course includes representing information with the relational database model, manipulating data with an interactive Structure Query Language (SQL) and database programming, database development including internet applications, and database security, integrity and privacy issues.

Course Contents:

Sessional on Database Systems: Concepts of database systems, Entity-Relationship concepts, Normalization, Relational model, SQL, Concurrency control, Recovery system, Advanced database management systems.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Address database questions and align the answers with the concepts of database management.		✓										
CLO2	Design SQL solutions for database manipulation and management.			✓									
CLO3	Apply modern tools for executing successful SQL commands to manipulate, recover and manage databases.					✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Tutorial • Discussion	• Quiz • Viva • Lab Report

		● Interaction ● Audio/Video ● Lab Manual	● Project ● Lab Final
CLO2	Cognitive (Create)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Lab Report ● Project ● Lab Final
CLO3	Cognitive (Apply)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Lab Report ● Project ● Lab Final

Course Code: CSE 3103

Course Title: Theory of Computation

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: There are several reasons why the study of automata and complexity is an important part of the core of Computer Science. Finite Automata are a useful model for many important kinds of hardware and software. After completing this course, students are expected to learn about software for designing and checking the behavior of digital circuits. They can explain the lexical analyzer of a typical compiler that is the compiler component that breaks the input text into logical units such as identifiers, keywords, and punctuations.

Course Contents:

Finite State Machine: Fundamental of Finite State Machine, State Equivalence and Minimization of Machine, Incompletely Specified Machine and Minimal Machine, Merger Graph and Compatibility Graph, Finite Memory and Definite Memory Machine, Information Lossless Machine and Inverse Machine.

Finite Automata: Introduction to Finite automata, Structural Representations, Automata and Complexity, an Informal Picture of Finite Automata, Deterministic Finite Automata, Nondeterministic Finite Automata, an Application (i.e. text search or other), Finite automata with Epsilon-Transitions.

Regular Expressions and Languages: Regular Expressions, Finite Automata and Regular Expression, Application of Regular Expressions, Algebraic Laws for Regular Expressions, Closure Properties of Regular Language, Decision Properties of Regular Languages, Equivalence and Minimization of Automata.

Context-Free Grammar and Languages: Context-Free Grammars, Parse Trees, Application of Context-Free Grammars, Ambiguity in Grammars and Languages, Normal Form for Context-Free Grammars, the Pumping Lemma for Context-Free Languages, Closure Properties of Context-Free Languages, Decision Properties of CFL's.

Pushdown Automata: Definition of the Pushdown Automata, the Languages of a PDA, Equivalence of PDA's and CFG's, Deterministic Pushdown Automata.

Introduction to Turing Machines: The Turing Machine, Programming Techniques for Turing Machines, Extensions to the Basic Turing Machine, Restricted Turing Machines, Turing Machines and Computers.

Undecidability: A language that is not recursively enumerable, an undecidable problem that is RE, undecidable problems about turing machines, post's correspondence problem, other undecidable problems.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Describing Finite State Machine and informal view of DFA and NFA for string processing.	✓											
CLO2	Understanding regular expressions and regular language.	✓											
CLO3	Analyzing the facts of context free grammars and languages and its automata.		✓										
CLO4	Interpreting the formation of a turing machine and undecidability.				✓								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Remember)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Psychomotor (Perception)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Presentation
CLO3	Cognitive (Analyze)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Project
CLO4	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Presentation

Course Code: CSE 3105

Course Title: Computer Interfacing and Embedded System

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: The course teaches interfacing of microcontrollers, microprocessors, and embedded system to the external world. This course will give an overview of computer interface components and their characteristics. It will cover detailed discussion on interfacing ICs, sensor, LED, LCD, seven segment display, transducers,

stepper motors, and peripheral devices. This course will enable students to develop theoretical and practical knowledge about embedded systems hardware as well as acquire skills in programming embedded systems.

Course Contents:

Microprocessor Interfacing: 8086/8088 Hardware Specifications, 8284A Clock Generator, Memory Interface, Address Decoding, 82C55 Programmable Peripheral Interface, 8254 Programmable Interval Timer, 8259A Programmable Interrupt Controller, 8237 DMA Controller, Co-processor, 80X87 Architecture, 80X87 Instruction Set, Programming with 80X87 Co-processor, 8251A Programmable Communication Interface.

Microcontroller Interfacing: Microcontroller Hardware Connection, Microcontroller Programming in C, HEX File, Flash Loader, I/O Port, Timer, Interrupt and Serial Port Programming, LCD, Keyboard, Display and Motor Interfacing, A/D Conversion, D/A Conversion and Sensor Programming, Waveform Generation, UART, SPI and I2C Protocol.

Embedded System: Overview of the Design Flow, Embedded Systems Specifications and Modeling, Embedded Hardware Platforms: Raspberry Pi & Arduino, Real Time Operating System (RTOS), Interfacing to the External World through Sensors and Actuators, Interfacing Ethernet, Bluetooth & 802.15.4.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding Microprocessor Chip and Its Interfacing with Memory Chip.	✓											
CLO2	Understanding Microprocessor Interfacing with I/O Devices.	✓											
CLO3	Understanding Microcontroller Programming.	✓											
CLO4	Understanding Microcontroller Interfacing.	✓											
CLO5	Designing Embedded System.			✓		✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Quiz • Viva • Class Test • Final Exam
CLO2	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Quiz • Viva • Class Test • Final Exam
CLO3	Cognitive (Understand)	• Lecture	• Quiz

		• Tutorial • Discussion • Interaction • Audio/Video	• Viva • Class Test • Final Exam
CLO4	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Quiz • Viva • Class Test • Final Exam
CLO5	Cognitive (Create)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Quiz • Viva • Class Test • Final Exam

Course Code: CSE 3106

Contact Hours/Week: 3/2 Hours

**Course Title: Computer Interfacing and Embedded System
Sessional**

Credit Hour: 0.75

Rationale: There are various reasons why one should take a course on peripherals and interfacing. This course is an essential component of the core of Computer Science. The Input Output Interface (I/O) is used to transfer information between internal storage and external I/O devices. Special communication links are required for peripherals connected to a computer to interface with the central processing unit. The communication link's aim is to bridge the gap between the central computer and each peripheral.

Course Contents:

Sessional on Computer Interfacing and Embedded System: Microprocessor Interfacing, Microcontroller Interfacing, Embedded System.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Applying Microprocessor Interfacing.		✓		✓								
CLO2	Applying Microcontroller Interfacing.		✓		✓								
CLO3	Designing Embedded System.			✓		✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction	• Quiz • Viva • Lab Report • Project

		• Audio/Video • Lab Manual	• Lab Final
CLO2	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO3	Cognitive (Create)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final

Course Code: CSE 3107
Course Title: Computer Architecture

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: This course aims to give an understanding of the mechanisms for implementing the programmer's idealized computer. It will start from Digital Logic Design course and will focus on implementing computer which can run assembly instructions taught at Assembly Language course. One important purpose of this course is to provide an in-depth understanding of how hardware and software interact with each other. This course will teach instruction sets, processor arithmetic and control, the Von Neumann architecture, pipelining, memory management, storage, and other input/output topics. The course will conclude with a look at the recent switch from sequential processing to parallel processing by looking at the other types of architectures.

Course Contents:

Introduction: Introduction to Von Neumann Organization, RISC and CISC Machines, HDL for Simple CPU Implementation, FPGA.

Data Representation and Computer Arithmetic: Big-Endian and Little-Endian Organization, Data Representation, Signed and Unsigned Formats, Computer Integer Arithmetic, Booth Multiplication Algorithm, IEEE Floating Point Representation and Arithmetic, FPU Design, Computer Arithmetic Circuits.

CPU Design: Fetch-Execute Cycle, Arithmetic and Logic Unit (ALU) and Register Set Design, CPU Instruction Set Design, Addressing Modes, Assembly Language and Assembler Design, Stack Implementation for Supporting Function Calling with Parameters, Interrupts, Exceptions, Operating System Calls and I/O Operations, Bus Architecture.

Memory Design: Memory Hierarchy, Memory Organization into Bytes and Words, Different Types of ROM and RAM Architectures, Flash Memory, Secondary Memory, Optical Memory, Read/Write Cycle, Cache Memory, Basic Cache Algorithm, Direct-Mapped and Associative Cache, Virtual Memory.

Pipelining and Parallel Processing: Pipelining, Pipeline Hazards, Branch Prediction, Speculative Execution, Instruction Level Parallelism, Superscalar Processor, Parallel Processing, Threading.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
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CLO1	Understanding basic computer architecture topics.	✓											
CLO2	Understanding different types of computer arithmetic operations, their representations, their implementation circuits and their HDL implementation.	✓											
CLO3	Analyzing different types of memory in memory hierarchy such as Cache Memory, Main Memory, Secondary Memory and Virtual Memory.		✓										
CLO4	Designing a complete Central Processing Unit (CPU) along with Control Unit (CU), Arithmetic & Logic Unit (ALU) & Instruction Set Architecture (ISA) and their HDL implementation.			✓									
CLO5	Implementing addressing modes in CPU, Assembler, Procedures and I/O operations.			✓									
CLO6	Understanding Pipelining and Parallel Processing Techniques.	✓											

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	Lecture	- Class Test - Final Exam
CLO2	Cognitive (Understand)	Lecture	- Class Test - Final Exam
CLO3	Cognitive (Analyze)	Lecture	- Class Test - Final Exam
CLO4	Cognitive (Create)	Lecture	Class Test

			Final Exam
CLO5	Cognitive (Create)	Lecture	- Class Test - Final Exam
CLO6	Cognitive (Understand)	Lecture	- Class Test - Final Exam

Course Code: CSE 3108

Contact Hours/Week: 3/2 Hours

Course Title: Computer Architecture Sessional

Credit Hour: 0.75

Rationale: This course aims to simulate a simple working CPU using Simulator. It also focuses on implementing the same CPU using Hardware Description Language (HDL).

Course Contents:

Sessional on Computer Architecture: Simple CPU and memory design using simulator and HDL.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Designing computer arithmetic circuits, arithmetic and logic unit (ALU) and register set.			✓	✓								
CLO2	Designing memory circuits.			✓	✓								
CLO3	Designing a CPU with Control Unit (CU) based Instruction Set Architecture (ISA) which supports addressing modes and pipelining.			✓	✓								
CLO4	Designing Assembler, Procedures, I/O implementation and System Call.			✓	✓								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Create)	- Lecture - Tutorial - Discussion	- Quiz - Viva - Lab Performance

		● Interaction ● Audio/Video ● Lab Manual	● Project ● Lab Final
CLO2	Cognitive (Create)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Lab Performance ● Project ● Lab Final
CLO3	Cognitive (Create)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Lab Performance ● Project ● Lab Final
CLO4	Cognitive (Create)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Lab Performance ● Project ● Lab Final

Course Code: CSE 3109

Course Title: Applied Statistics and Queuing Theory

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: Statistics and probability are the integral part of academic studies, scientific industry and most importantly research fields. Collection, analysis and extraction of knowledge from data are the preamble of any research-based studies. Server-client is a concept on which the modern internet is built on. Quality of service, optimization in resource allocation and concept of queuing theory are playing crucial role by transparentizing to end users and making the service access more enjoyable. Anyone with sufficient knowledge in this course will be able to create positive impacts on the society by discovering unprecedented knowledge.

Course Contents:

Introduction: Statistics and its Importance, Population and Sample, Variable and Constants, Statistical Data, Data Collection and Presentation, Construction of Frequency Distribution and Graphical Presentation.

Measures of Central Tendency: Arithmetic Mean, Geometric Mean, Harmonic Mean, Median, Mode, Weighted Mean.

Measures of Dispersion: Range, Standard Deviation, Variance, Moments, Skewness and Kurtosis.

Correlation Theory and Causality Analysis: Linear Correlation and its Measures and Significance, Rank Correlation, Causality Analysis.

Tests of Significance: Hypothesis Testing, P-value, t-test, z-test, Statistical Significance, Statistical Inference.

Probability: Elementary Concepts, Laws of Probability – Additive and Multiplicative Law, Conditional Probability and Bay's Theorem, Random Variables, Mathematical Expectation.

Probability Distributions: Binomial Distribution, Poisson Distribution and Normal Distribution.

Queuing Theory: Stochastic Processes, Discrete Time Markov Chain and Continuous Time Markov Chain. Birth-Death Process in Queuing. Queuing Models: M/M/1, M/M/C, M/G/1, M/D/1, G/M/1 Solution of Network of Queue-Closed Queuing Models and Approximate Models. Application of Queuing Models in Computer Science.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding fundamentals of statistics and statistical data		✓										
CLO2	Applying Central Tendency, and Dispersion on real data.				✓								
CLO3	Applying correlation, causality and hypothesis testing.				✓								
CLO4	Understanding the elementary concept of probability.		✓										
CLO5	Applying probability distributions in problem solving.			✓									
CLO6	Applying queuing models in various applications and problem solving.			✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Affective (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO3	Affective (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO4	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO5	Affective (Apply)	• Lecture • Discussion	• Class Test • Final Exam

		• Interaction	
CL06	Affective (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam

3rd YEAR 2nd SEMESTER

Course Code: CSE 3200

Course Title: Software Development Project II

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: This course will enable the student to go through different phases of the Software development cycle and produce a higher quality software as an end product. This will also help them to get an idea of different types of database systems and how to provide proper documentation for software.

Course Contents:

Students will Work in Groups or Individually to Develop High Quality Software/Projects Including New I/O Drivers or Similar Projects Involving Operating Systems Modules in Different Types of Database Systems or Project Oriented and Visual Languages, Students will Writes Structure Program and use Proper Documentation.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understand the steps of Software Development Life Cycle and its practical implementation.		✓		✓								
CLO2	Differentiate between different types of database systems and its proper usage					✓							
CLO3	Create code repository and proper documentation.			✓						✓			
CLO4	Create prototypes and final versions of software.			✓						✓		✓	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Viva • Project
CLO2	Cognitive (Analyze)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Viva • Project
CLO3	Cognitive (Create)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Viva • Project
CLO4	Cognitive (Create)	• Lecture • Tutorial	• Viva • Project

	systems interact with computer hardware.												
CLO4	Developing different structures and mechanisms of operating systems to overcome different complexities.			✓		✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO2	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO3	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO4	Cognitive (Create)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment

Course Code: CSE 3202

Course Title: Operating System Sessional

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: Operating systems are central to any computing device. This course is designed to introduce the internal operations and structures of modern operating systems. This course will cover topics which will help students to understand the components of operating systems. At the end of the course students will be able to explain how a modern operating system interacts with the computer hardware and apply the concepts to design solutions for different practical problems.

Course Contents:

Sessional on Operating System: Shell Programming, System Programming, OS Development.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understand operating an operating system from the command line.	✓											✓
CLO2	Understand how the components of modern operating systems interact with computer hardware.			✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO2	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final

Course Code: CSE 3203

Course Title: Data Communication

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: To effectively communicate between the devices, it is critical to precisely play out every essential apparatus and its fundamental design. Therefore, comprehending the system structure, segments and other applicable devices and procedures is an absolute necessity. This course will provide a prologue to the field data communications, Layered design of LAN, MAN and WAN, fundamentals of data and signals, basic transmission concepts, basic of transmission medium, transmission error detection and correction etc.

Course Contents:

Fundamental: Representation of signals in time and frequency domain, Properties of Fourier transform, Auto-correlation, Cross-correlation.

Data Communication and Network Model: Data communication, Fundamentals of networks, History of the internet, Protocols and standards.

Signal and System, Transmission Media, Interfaces: Analog and digital data, Periodic analog signals, Digital signals, Transmission impairment, Data rate limits and performance.

Analog Transmission: Digital-to-Analog conversion: ASK, FSK, PSK, QAM. Analog-to-Analog conversion: Amplitude modulation, Frequency modulation, Phase modulation.

Digital Transmission: Digital-to-Digital conversion: Line coding, block coding, scrambling, Analog-to-Digital conversion: Pulse code modulation, Delta modulation, Transmission modes: Parallel and serial transmission.

Multiplexing, Spreading and Switching: Bandwidth Utilization, Multiplexing, Spread spectrum, Packet-switched data networks, Circuit-switched data networks, Virtual packet networks.

Transmission Medium: Guided and unguided media.

Error Detection and Correction: Data word, Codeword, Redundancy, Forward Error Detection, Retransmission, Parity Checker, Single Bit Error, Burst Errors, LRC, VRC, CRC, Checksum, Hamming Code.

Telephone Networks: Telephone Networks, Dial-up Modems, and Digital Subscriber Line.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
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CLO1	Understanding the basic terminologies of data communication and various types of data.		✓										
CLO2	Analyzing the performance and implementation of analog transmission techniques.			✓									
CLO2	Analyzing the performance and implementation of digital transmission techniques and telephone networks.			✓									
CLO3	Illustrating various types of multiplexing, spread spectrum and switching techniques.			✓									
CLO5	Understanding various transmission mediums and applying error detection and correction techniques to detect and correct the errors in data transmission.		✓										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Analyze)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO3	Cognitive (Analyze)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO4	Cognitive (Remember)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO5	Cognitive (Understand & Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam

Course Code: CSE 3204

Course Title: Data Communication Sessional

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: The objective of this course is to cover the principles of analog and digital communication systems involving different modulation and coding schemes. This course will enable students to understand how these systems work and have an ability to apply knowledge to real-world problems.

Course Contents:

Sessional on Data Communication: Introduction to Tuned Circuits, Modulator and Demodulator, Multiplexer and Demultiplexer.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding different electronics components underlying data communication		✓										
CLO2	Analyzing implementation of analog and digital transmission techniques.					✓							
CLO3	Analyzing implementation of multiplexing techniques.					✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Lab Report ● Project ● Lab Final
CLO2	Affective (Analyze)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Lab Report ● Project ● Lab Final
CLO3	Affective (Analyze)	● Lecture ● Tutorial ● Discussion ● Interaction ● Audio/Video ● Lab Manual	● Quiz ● Viva ● Lab Report ● Project ● Lab Final

Course Code: CSE 3205

Course Title: Software Engineering

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: Software Engineering is about the discipline needed to develop high-quality software that can be understood, maintained, and adapted over long periods by many different people. This course will introduce the different software process models, project management, software requirements engineering process, systems analysis and design as a problem-solving activity, key elements of analysis and different design patterns, and the place of the analysis and design phases within the system development life cycle.

Course Contents:

Introduction: Introduction to Software and its Nature, Software Engineering Methods, Professional and Ethical Responsibility of a Software Engineer.

Software Process Model: Different Types of Software Process Model and their Implementations, Agile Software Process, Costs of Software Engineering.

Software Requirement Analysis: Software Requirements Analysis and their Applications, Software Prototyping, Basic Concepts of Different Formal Software Specification.

Design of Software: Software Design and its Different Techniques, Version Control, Software Configuration Managements. System Structuring, Control Models, Modular Decomposition, Domain-Specific Architecture.

Design Patterns: Introduction to design patterns, Creational Design Patterns: Factory, Singleton, Object Pool, Prototype, Builder, Structural Design Patterns: Adapter, Composite, Façade, Proxy, Decorator, Behavioral Design Patterns: Strategy, Observer, Command, Template Method, Iterator.

Software Testing: Software Validation and Verification: Verification and Validation Planning, Software's Testing Strategies and Different Type of Testing Techniques, Art of Debugging.

Software Quality Assurance: Software Quality Assurance Activities, Software Quality Assurance Plan, Software Cognitive Fundamentals, Concepts of Software Reengineering and Web Engineering.

Advanced Topics: Software Reliability Metrics, Software Reliability Specification, Statistical Testing and Reliability Growth Modeling, Use of CASE Tools and Technological Support in Engineering Software, Introduction to Unified Modeling Language–UML.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

[illegible]

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam
CLO2	Cognitive (Apply)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam
CLO3	Cognitive (Create)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam
CLO4	Cognitive (Understand & Apply)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam
CLO5	Cognitive (Analyze)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam
CLO6	Cognitive (Understand)	● Lecture ● Discussion ● Interaction	● Class Test ● Final Exam

Course Code: CSE 3206

Course Title: Software Engineering Sessional

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: The objective of this course is to have students hands-on experience on practical aspects of software engineering. This course includes different design patterns, different software testing methods and software configuration management tools. This course will help students get better at developing real-life software.

Course Contents:

Sessional on Software Engineering: Software Configuration Management Tool, Design Patterns, Software Testing.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

[illegible]

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Project • Lab Report • Lab Final
CLO2	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Project • Lab Report • Lab Final
CLO3	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Project • Lab Report • Lab Final

Course Code: CSE 3207**Course Title: Artificial Intelligence****Contact Hours/Week: 3 Hours****Credit Hour: 3.00**

Rationale: Artificial Intelligence (AI) is one of the hot topics in the Computer Science and Engineering field in present day. It is at the heart of current technology and the applications and understanding of AI helps to solve many contemporary and future real-world problems. Artificial intelligence is a research domain and the knowledge of AI helps us to understand the intelligent human behaviors on a computer. The ultimate key features of AI are to make a computer in such a way that it can plan, learn, and resolve problems autonomously. The main research topics in artificial intelligence include reasoning, general problem-solving approaches, natural language processing and so on.

Course Contents:

Fundamental: Definition of AI, historical development of AI, application of AI.

Production systems: Introduction of product system, production rules, types of production systems, knowledge base optimization in a production system.

General Problem Solving Approaches: Breadth first search, depth first search, iterative deepening search, hill climbing, simulated annealing, heuristic search, A* algorithm, adversary search, the minimax algorithm, constraint satisfaction problems, Genetic Algorithm: Basic concepts, offspring, reproduction, selection, crossover, mutation, fitness.

Logic and Structural Knowledge Representation: Propositional logic, first-order logic, resolution principle, frames, semantic-nets, petri nets, relational data model.

Reasoning under Uncertainty: Bayesian reasoning, fuzzy knowledge, probability theory, fuzzy set theory, expert systems.

Natural language processing: Naive Bayes algorithm, syntactic semantics and pragmatic, lexicon.

Programming Languages for AI Research: Historical overview, features of AI programming languages, Implementation of AI algorithms through Python.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding the basic knowledge of artificial intelligence in computer science and engineering	✓											
CLO2	Applying different search techniques for solving real world problems		✓										
CLO3	Explaining structural knowledge and reasoning under uncertainty		✓				✓						
CLO4	Analyzing different natural language processing approaches.		✓	✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO3	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO4	Cognitive (Analyze)	• Lecture • Discussion • Interaction	• Class Test • Final Exam

Course Code: CSE 3208**Course Title: Artificial Intelligence Sessional****Contact Hours/Week: 3/2 Hours****Credit Hour: 0.75**

Rationale: In this course, theoretical knowledge of the course CSE 3207 is implemented in the laboratory. Lab experiments are conducted according to the theory course. At the end of this course, students are able to design and implement different search techniques, structural knowledge, reasoning, and natural language processing approaches using Python.

Course Contents:

Sessional on Artificial Intelligence: Python Programming, General Problem Solving Approaches, Logic and Structural Knowledge Representation, Reasoning under Uncertainty, Natural Language Processing.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
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CLO1	Understanding the application of Python for solving AI problems.					✓							
CLO2	Designing and implementing different search techniques using Python.		✓			✓							
CLO3	Designing and implementing knowledge based systems using Python.		✓			✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Performance • Project • Lab Final
CLO2	Cognitive (Create)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Performance • Project • Lab Final
CLO3	Cognitive (Create)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Performance • Project • Lab Final

Course Code: CSE 3209

Course Title: Digital Signal Processing

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: All real-world data is analog, and these data are transformed or communicated from one place to another through digital devices, whereas the digital devices use digital signal processing system for storing the data. Moreover, digital signal processing (DSP) can analyze, synthesize, modify, separate, enhance, and modify various audio, image, video, and communication signals.

Course Contents:

Introduction: Signals, Systems and Signal Processing, Classification of Signals, the Concept of Frequency in Continuous Time and Discrete Time Signals, Analog to Digital and Digital to Analog Conversion, Sampling and Quantization, Under Sampling and Oversampling of a digital Signal.

Discrete Time Signals and Systems: Discrete Time Signals, Discrete Time Systems, Analysis of Discrete Time Linear Time Invariant Systems. Discrete Time Systems Described by Difference Equations, Implementation of Discrete Time Systems, Correlation and Convolution of Discrete Time Signals.

The Z-Transform: Introduction, Definition of the Z-Transform, Z-Transform and ROC of Infinite Duration Sequence, Properties of Z-Transform Inversion of the Z-Transform, the One-Sided Z Transform.

Frequency Analysis of Signals and Systems: Frequency Analysis of Continuous Time Signals, Frequency Analysis of Discrete Time Signals, Properties of Fourier Transform of Discrete Time Signals, Frequency Domain Characteristics of Linear Time Invariant System, Linear Time Invariant Systems as Frequency Selective Filters, Inverse Systems and De-convolution.

Discrete Fourier Transform (DFT): Discrete Fourier Series (DFS), Properties of DFS, Discrete Fourier Transformation (DFT), Properties and Application of DFT.

Fast Fourier Transform (FFT): FFT Algorithms, Applications of FFT Algorithm.

Digital Filter Design: Differential and Difference Equations, Digital Transfer Functions, Frequency Response, Digital Filter Realization Scheme, Finite Impulse Response (FIR) Infinite Impulse Response (IIR) Filter Design.

Application of DSP: Speech, Image and Video Processing.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding the underlying concepts of system and basic signal processing.	✓											
CLO2	Applying knowledge of Fourier transform and Z-transform to acquire information about various LTI systems and filters.	✓											
CLO3	Analyzing the spectra of linear time invariant system (LTI) and various discrete transforms like DFT and FFT.		✓										
CLO4	Developing digital filters and digital signal processing (DSP) system.			✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Apply)	• Lecture • Discussion • Interaction • Audio/Video	• Class Test • Final Exam • Assignment
CLO3	Cognitive (Analyze)	• Lecture • Discussion • Interaction • Audio/Video	• Class Test • Final Exam
CLO4	Cognitive (Create)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment

			● Presentation
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Course Code: CSE 3210

Course Title: Digital Signal Processing Sessional

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: In this course, theoretical knowledge is implemented in the laboratory based on the course CSE – 3209 (Digital Signal Processing). Lab experiments are designed according to the theory course. At the end of this course students can design various linear time invariant systems (LTI) and various filters. Students are also able to analyze various digital signal processing applications such as image & video compression concepts, speech processing etc.

Course Contents:

Sessional on Digital Signal Processing: LTI System, Z Transform, Frequency Analysis of Signals and Systems, Discrete Fourier Transform (DFT), Fast Fourier Transform (FFT), Filter Design.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Implementing various signal processing systems by using or applying Mathematical knowledge.	✓	✓										
CLO2	Analyzing the spectrum of various signals by using the knowledge of various transforms.	✓	✓										
CLO3	Developing a digital signal processing system for real life applications such image and video coding.			✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Apply)	● Lecture ● Tutorial ● Discussion ● Interaction ● Lab Manual	● Quiz ● Viva ● Lab Report ● Presentation ● Project ● Lab Final

CLO2	Cognitive (Analyze)	● Lecture ● Discussion ● Interaction ● Lab Manual	● Quiz ● Viva ● Lab Report ● Project ● Lab Final
CLO3	Cognitive (Create)	● Lecture ● Discussion ● Interaction ● Lab Manual	● Quiz ● Viva ● Lab Report ● Project ● Lab Final ● Presentation

4th YEAR 1st SEMESTER

Course Code: CSE 4000
Course Title: Project/Thesis I

Contact Hours/Week: 2 Hours
Credit Hour: 1.00

Rationale: A thesis is a document submitted in support of candidature for an academic degree or professional qualification presenting the students' research and findings. On the other hand, a project is any undertaking, carried out individually or collaboratively and possibly involving research or design, that is carefully planned to achieve a particular aim. In this course, students can choose any one of these two.

Course Contents:

Study of Problems in the Field of Computer Science and Engineering. The Project/Thesis Topic Selected in this course will be continued in the CSE 4000 course of 4th year 2nd semester.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

[illegible]

	and manage the project/thesis in a manner so that the solution can be kept feasible.												
CLO10	Utilize the designed solutions and publications to create career opportunities.												✓

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Create)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CLO2	Cognitive (Evaluate)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CLO3	Cognitive (Apply)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CLO4	Affective (Internalize)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CLO5	Psychomotor (Articulate)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CL06	Affective (Value)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CL07	Affective (Organize)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CL08	Affective (Respond)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CL09	Cognitive (Evaluate)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CL10	Psychomotor (Embody)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense

Course Code: CSE 4101
Course Title: Compiler Design

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: Compiler design introduces the theory and tools that can be employed in order to perform syntax-directed translation of a high-level programming language into an executable code. This course introduces regular languages and context free languages which are crucial to understand how compilers and programming languages are built. These techniques can also be employed in wider areas of applications whenever we need a syntax-directed analysis of symbolic expressions and languages and their translation into a lower-level description.

Course Contents:

Introduction to Compiler: Compiler Structure, Analysis-Synthesis Model of Compilation, Various Phases of a Compiler, Tool based Approach to Compiler Construction, Compiler-Compilers and Translator Writing Systems, Cross-Compilation, Just-In-Time Compilation, Ahead-Of-Time Compilation, Bootstrapping, Source-to-Source Compiler.

Lexical Analysis: Interface with Input, Parser and Symbol Table, Token, Lexeme and Patterns, Difficulties in Lexical Analysis, Error Reporting, Implementation, Regular Definition, Transition Diagrams, Lex.

Syntax Analysis: Top-Down Parsing, Recursive Descent Parsing, Transformation on the Grammars, Predictive Parsing, Bottom-Up Parsing, Operator Precedence Grammars, LR Parsers (SLR, LALR, LR), YACC.

Syntax Directed Translation: Inherited and Synthesized Attributes, Dependency Graph, Evaluation Order, Bottom Up and Top Down Evaluation of Attributes, L- and S-Attributed Definitions.

Type Checking: Type System, Type Expressions, Structural and Name Equivalence of Types, Type Conversion, Overloaded Functions and Operators, Polymorphic Functions.

Run Time System: Storage Organization, Activation Tree, Activation Record, Parameter Passing, Dynamic Storage Allocation, and Heap Storage Management, Garbage Collection.

Intermediate Code Generation: Intermediate Representations, Translation of Declarations, Assignments, Control Flow, Boolean Expressions and Procedure Calls, Implementation Issues.

Code Generation and Instruction Selection: Issues, Basic Blocks and Flow Graphs, Register Allocation, Code Generation, Dag Representation of Programs, Code Generation from Dags, Peep Hole Optimization, Code Generator Generators, Specifications of Machine, Code Optimization Techniques.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding internals of compiler and the mechanism of lexical analyzer for generating tokens.		✓										
CLO2	Understanding syntax analyzer to use tokens for accurate parse tree generation.		✓										
CLO3	Applying syntax directed definition and syntax directed translation to build type system and syntax tree.			✓									
CLO4	Analyzing different types of intermediate code representations.			✓									
CLO5	Evaluating different storage allocation				✓								

	of how a compiler works.												
CLO2	Understanding how LEX and YACC is used for lexical and syntax analysis.				✓								
CLO3	Designing front end of compiler (Lexical Analysis, Syntax Analysis and Semantic Analysis) using LEX and YACC.			✓									
CLO4	Designing back end of compiler (Intermediate Code Generation and Code Generation) using LEX and YACC.			✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Lab Performance • Project • Lab Final
CLO2	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Lab Performance • Project • Lab Final
CLO3	Cognitive (Create)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Lab Performance • Project • Lab Final
CLO4	Cognitive (Create)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Lab Performance • Project • Lab Final

Course Code: CSE 4103
Course Title: Computer Networks

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: Computer networking has become one of the most successful ways of sharing information, where all computers are linked together by a common network. Students who are Computer Science and Engineering major need to enhance the skills of building up a wired or wireless network and sharing the resources among the stakeholders including individuals, offices, and community.

Course Contents:

Introduction: Definition, Uses of Computer Networks, Network Topology, Network Media, Network Devices, Different Types of Networks: PAN, LAN, MAN, WAN etc.

Network Model: OSI Reference Model, TCP/IP Reference Model, Functions of the Layers of Different Models, Network Protocols Working at Different Layers.

IP Addressing: IP Addressing, IPV4 & IPV6, Classification of IP Addressing, Subnet Mask, CIDR, Private IP Address, Public IP Address, Sub Netting, VLSM etc.

Data Link Layer: Data Link layer Design Issues and Framing: Character Count, Byte Stuffing, Bit Stuffing.

Data Link Layer Protocols: Simplex Stop & Wait Protocol, Piggybacking, Point to Point Protocol (PPP), Sliding Window Protocols (One Bit Sliding Window Protocol, Go Back N ARQ, Selective Repeat ARQ).

Media Access Control: Random Access: ALOHA, CSMA, CSMA/CD, CSMA/CA; Controlled Access: Reservation, Poling, Token Passing; Channelized Access: CDMA, TDMA, FDMA.

Wired & Wireless LANs: IEEE 802. *, Ethernet, Sonet, ATM, ATM, Bluetooth, WiMAX.

Network Layer: Network Layer Design Issues, Various Routing Algorithms (Static & Dynamic Routing, Optimality Principle, Shortest Path Routing, Flooding, Distance Vector Routing, Link State Routing, Hierarchical Routing, Multicast & Broadcast Routing etc.), Congestion Control Algorithms, Quality of Service, Internetworking, Internetworking Devices etc.

Network Layer Protocols: Address Resolution Protocol, Internet Protocol, Internet Control Message Protocol, Routing Information Protocol, Open Shortest Path First, Border Gateway Protocol, User Datagram Protocol, Transmission Control Protocol.

Transport Layer: Elements of Transport Protocols; Connection Establishment (Three Way Handshake), Connection Release (Blue & White Army Problem).

Application Layer & Others: HTTP and Recent Advances in Internet Protocols, Web Server Performance, Proxy Servers, Load Balancing in Web Servers, IP Security, Queuing Models for Networks and Protocols.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

[illegible]

	wired, and wireless LANs standard.												
CLO5	Analyzing various routing algorithms of network layer.		✓										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO3	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO4	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO5	Cognitive (Analyze)	• Lecture • Discussion • Interaction • Audio/Video	• Class Test • Final Exam

Course Code: CSE 4104

Course Title: Computer Networks Sessional

Contact Hours/Week: 3 Hours

Credit Hour: 1.50

Rationale: Computer networking has become one of the most successful ways of sharing information, where all computers are linked together by a common network. Students who are Computer Science and Engineering major need to enhance the skills of building up a wired or wireless network and sharing the resources by providing network security among the stakeholders including individuals, offices and community.

Course Contents:

Sessional on Computer Networks: Introduction to CISCO Packet Tracer and NS2 (Network Simulation Version 2), IP Addressing, Routing Algorithms, Socket Programming.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Configuring IP addressing technologies including Sub-netting and VLSM into a computer network.					✓		✓					
CLO2	Configuring various routing algorithms					✓		✓					
CLO3	Applying socket programming to develop network applications					✓		✓					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO2	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO3	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final

Course Code: CSE 4105**Course Title: Digital Image Processing****Contact Hours/Week: 3 Hours****Credit Hour: 3.00**

Rationale: This course is designed to give undergraduate level fundamental knowledge in digital image processing. The course will cover basic theory and principles of digital image processing with emphasis in image processing techniques, image filtering design and applications. It will extend the signals and systems knowledge of the students to two-dimensional signals. This is a very important course for any student who wants to do thesis related to computer vision.

Course Contents:

Digital Image Fundamentals: Different Types of Digital Images, Sampling and Quantization, Imaging Geometry, Image Acquisition Systems.

Bilevel Image Processing: Basic Concepts of Digital Distances, Distance Transform, Medial Axis Transform, Component Labeling, Thinning, Morphological Processing, Extension to Grayscale Morphology.

Binarization of Grey Level Images: Histogram of Grey Level Images, Optimal Thresholding using Bayesian Classification, Multilevel Thresholding.

Edge Detection: First Order and Second Order Edge Operators, Multi-Scale Edge Detection, Canny's Edge Detection Algorithm, Hough Transform for Detecting Lines and Curves, Edge Linking.

Image Enhancement: Point Processing, Spatial Filtering, Frequency Domain Filtering, Multi-Spectral Image Enhancement, Image Restoration.

Image Segmentation: Segmentation of Grey Level Images, Water Shade Algorithm for Segmenting Grey Level Image. Image Representation and Description, Recognition and Interpretation.

Image Compression: Lossy and Lossless Compression Schemes, Prediction Based Compression Schemes, Vector Quantization, Sub-band Encoding Schemes, JPEG Compression Standard, Fractal Compression Scheme, Wavelet Compression Scheme.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Have a clear understanding of digital image representation including reading, writing and manipulation in computer with mathematical formulation.	✓	✓										
CLO2	Develop a theoretical foundation of fundamental digital image processing concepts like preprocessing, filtering/ masking, and transformation.		✓	✓									
CLO3	Combine existing image analysis techniques to solve emerging problems like segmentation, clustering, classification or compression.				✓	✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Affective (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Presentation
CLO3	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Project

Course Code: CSE 4106**Course Title: Digital Image Processing Sessional****Contact Hours/Week: 3/2 Hours****Credit Hour: 0.75**

Rationale: This laboratory is developed on the course designed to cover basic theory and principles of digital image processing techniques. The laboratory will give emphasis in image processing techniques like basic point processing, image filtering in spatial/frequency domain and the applications in image enhancement or other related applications. The overall knowledge will be used in some high level image processing approach like segmentation, compression, clustering and others.

Course Contents:

Sessional on Digital Image Processing: Digital Image Fundamentals, Bilevel Image Processing, Binarization of Grey Level Images, Edge Detection, Image Enhancement, Image Segmentation, Image Compression.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Demonstrate basic operations for image representation and point processing.	✓	✓										
CLO2	Examine the performance of spatial domain and frequency domain masking/filtering operations for image enrichment and other purposes.		✓	✓									
CLO3	Combine existing image analysis techniques to solve application problems.				✓	✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO2	Affective (Respond)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO3	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final

Course Code: CSE 4108

Course Title: Industrial Attachment

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: This is a students' industrial work experience course that serves to impart valuable practical experiences that prepare students for the workforce upon graduation.

Course Contents:

Students will visit industries and collaborate with them via interactive sessions for a specific period defined by the corresponding industry and the department.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Analyze, investigate and		✓	✓	✓								

	design solution modules for the assigned tasks for the industries.												
CLO2	Learn the usage of modern industrial tools in the domains of technology.					✓							
CLO3	Learn about the impact if industry on society, environment and sustainability.						✓	✓					
CLO4	Understand the work ethics and effectively communicate with the team as a team member/ leader.								✓	✓	✓		
CLO5	Embody the concepts of project management and finance to generate effective solutions in future.											✓	✓

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Apply, Analyze, Evaluate, Create)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Cont. Asses. • Viva
CLO2	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Cont. Asses. • Viva
CLO3	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Cont. Asses. • Viva
CLO4	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Cont. Asses. • Viva
CLO5	Psychomotor (Embody)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Cont. Asses. • Viva

Rationale: The aim of this course is to understand the need for a professional approach to develop an information system. Upon completion the course, the student should be able to: (i) describe principles, concepts and practice of information system analysis and design process (ii) explain the process of constructing different types of information system (iii) apply the design concepts to capture a business requirements (iv) design and develop of information systems in various real-world environment

Course Contents:

Application Development Policy and Strategies: Planning of Information System, Policy in Information System Development, Strategies for Achieving Information System Goals.

Application System Development Life Cycle: Phases in Application System Development, Interrelationship among Each Phase.

Feasibility Assessment: Problems and Needs in Information System Development, Preliminary Application Requirements Determination, Feasibility Assessment: Economic, Technical, Operational and Schedule Feasibility.

Information Requirements Determination: Strategies for Obtaining Information Requirements, Technique for Information Requirements Determination, Methods for Providing Assurance that Requirements are Correct and Complete.

Structured Systems Analysis: Steps in Structured Systems Analysis, Activity Diagrams and Related Documentation, Data Dictionary, Problem Analysis, Structured Walk Through.

Systems Design Methodology: Check List Methodology, Process-Oriented Methodology, Application Generator, Structured Design.

Program Development and Testing: Structured Programming and Method for Testing.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding and analyzing application development life cycle and strategies		✓	✓		✓				✓			
CLO2	Applying feasibility assessment and determining information requirements			✓	✓	✓							
CLO3	Understanding system design methodology and testing and analyzing structured system						✓	✓				✓	✓

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand & Analyze)	- Lecture - Discussion - Interaction	- Class Test - Final Exam

CLO2	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO3	Cognitive (Understand & Analyze)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment

Course Code: CSE 4110

Contact Hours/Week: 3/2 Hours

Course Title: Information Systems Analysis and Design Sessional

Credit Hour: 0.75

Rationale: The aim of this course is the development of student's technical and interpersonal skills as a system analyst. After completing this course students should be able to design and develop information systems in various real-world environments.

Course Contents:

Sessional on Information Systems Analysis and Design: Identification of the problems, Initial investigation, Detailed Analysis of the Information System, Design alternative solutions of the identified problems, Finalize the design of the targeted information system.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Analysis of the information system.		✓		✓				✓	✓	✓		
CLO2	Design of the information system.			✓		✓	✓	✓		✓			
CLO3	Solve real life complex problems.		✓	✓	✓	✓	✓	✓		✓	✓	✓	✓

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Analyze) Affective (Organize)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Presentation • Project • Lab Final
CLO2	Cognitive (Create and Design)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Presentation • Project • Lab Final
CLO3	Cognitive (Create and Design), Psychomotor (Response) Affective (Value)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Presentation • Project • Lab Final

Course Code: CSE 4111
Course Title: Unix Programming

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: The primary objective of this course is to introduce the UNIX Operating System and its File System along with an understanding of important aspects related to the SHELL and the process. It will focus on developing students' ability to formulate regular expressions and use them for pattern matching and provide a comprehensive introduction to SHELL programming, services and utilities. The specific topics include, but not limited to basic commands, process status, handling Unix signals, permissions, piping, multi-threading, regular expressions, IPC, shell scripting.

Course Contents:

Introduction: Introduction to Unix Programming.

Unix Environment: Command Line, Globbing, I/O Redirection, Piping, Basic Commands, Memory Layout.

Debugging: GDB, Valgrind, Essential X86, Fork, Exec, Wait, Process Status, Bit Manipulation, Sending Signals Unix I/O Implementing I/O Redirection, Piping Directories and Files. Walking a Directory Tree, Exploring Attributes. Implementing Ls-L. Permissions, File Owner / Group, Time-Stamps. Signals and Signal Handling Design / Implementation of Sleep Process Relationships Backgrounding. Popen / Pclose Midterm Terminal Handling Review Midterm Networking Client / Server. I/O Multiplexing.

Multi-threading: Basics, Mutual Exclusion Multi-Threading: Bounded Buffers, Condition Variables Multi-Threading: Deadlocks Non-Blocking I/O. Regular Expressions. Sys V IPC. Semaphores and Shared Memory. Shell Scripting.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Describing the architecture and features of UNIX Operating System and distinguish it from other Operating System	✓			✓								
CLO2	Demonstrating UNIX commands for file handling and process control	✓			✓								
CLO3	Writing Regular expressions for pattern matching and apply them to various filters for a specific task	✓			✓					✓			
CLO4	Analyzing a given problem and apply requisite facets of SHELL programming in order to devise a SHELL script to solve the problem	✓			✓					✓			

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	- Lecture - Discussion	- Class Test - Final Exam

			• Assignment
CLO2	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO3	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction	• Class Test • Final Exam • Project • Presentation
CLO4	Cognitive (Apply, Analyze)	• Lecture • Tutorial • Discussion • Interaction	• Class Test • Final Exam • Project • Presentation

Course Code: CSE 4112

Course Title: Unix Programming Sessional

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: This course introduces basic understanding of UNIX OS, UNIX commands and File system and to familiarize students with the Linux environment. To make students learn fundamentals of shell scripting and shell programming. Emphases are on making students familiar with the UNIX environment and issues related to it.

Course Contents:

Sessional on Unix Programming: Unix Environment, Debugging, Multi-threading.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Run various UNIX commands on a standard UNIX/LINUX Operating system	✓	✓	✓		✓							
CLO2	Do shell programming on UNIX OS	✓	✓	✓		✓							
CLO3	Understand and handle UNIX system calls	✓	✓	✓		✓				✓			

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand, Analyze, Apply)	• Lecture • Tutorial • Discussion	• Quiz • Viva • Lab Report • Project • Lab Final
CLO2	Cognitive (Analyze, Apply)	• Lecture • Tutorial • Interaction	• Quiz • Viva • Lab Report • Project • Lab Final
CLO3	Cognitive (Understand, Analyze, Apply, Create)	• Lecture • Tutorial	• Quiz • Viva

	Affective (Receive, Respond)	• Discussion • Interaction	• Lab Report • Project • Lab Final
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Course Code: CSE 4113

Course Title: Digital System Design

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: There are several reasons why the study of Digital System Design is an important part of the core of Computer Science. The students can attain the practical knowledge of designing I/O systems and I/O devices and the microprocessor design process. They are able to develop programmable peripherals interface such as A/D and D/A converters and interrupt systems. It helps them to attain knowledge on designing different sub systems of computer such as ALU, RAM, Control Units etc.

Course Contents:

System Design: Designing I/O System; I/O Devices; Designing Microprocessor based System with Interfacing Chips.

Programmable Design: Programmable Peripheral Interface (Interface to A/D and D/A Converter); Keyboard/Display Interface; Programmable Timer; Programmable Interrupt Controller, DMA Controller.

Memory Design: Design using MSI and LSI Components; Design of Memory Subsystem using SRAM and DRAM.

Design of Various Components of a Computer: ALU, Memory and Control Unit, Hardwired and Micro Programmed; Microprocessor based Designs; Computer BUS Standards; Design Special Purpose Controllers.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Describing the design process of I/O system; I/O devices.	✓		✓									
CLO2	Describe the steps of designing programmable Peripheral Interface.	✓	✓			✓							
CLO3	Developing ALU, Memory and Control Unit etc.		✓			✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Remember)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Psychomotor (Embody)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Presentation
CLO3	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Project

Course Code: CSE 4114
Course Title: Digital System Design Sessional

Contact Hours/Week: 3/2 Hours
Credit Hour: 0.75

Rationale: This course motivates students to learn about the way in which various computer components are constructed. This will also help the student to program several types of peripheral interface and to design different kinds of memory systems.

Course Contents:

Sessional on Digital System Design: Designing I/O systems and I/O devices, Designing MSI and LSI components and SRAM, DRAM, Designing ALU, Memory and Control Unit.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Designing different I/O systems and devices and Programming different Peripheral Interface.			✓									
CLO2	Understand the design process of Memory Subsystem and different embedded integrating System.				✓								
CLO3	Have a clear understanding about construction process of different computer components.		✓		✓								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Create)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO2	Psychomotor (Manipulate)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO3	Cognitive (Analyze)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final

Course Code: CSE 4115
Course Title: Simulation and Modeling

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: Simulation modeling solves real-world problems safely and efficiently. It provides an important method of analysis which is easily verified, communicated, and understood. Across industries and disciplines, simulation modeling provides valuable solutions by giving clear insights into complex systems.

Course Contents:

Simulation Modeling Basics: Systems, Models and Simulation; Classification of Simulation Models; Steps in a Simulation Study.

Concepts in Discrete-Event Simulation: Event-Scheduling vs. Process-Interaction Approaches, Time-Advance Mechanism, Organization of a Discrete-Event Simulation Model; Continuous Simulation Models; Combined Discrete-continuous models; Monte Carlo Simulation; Simulation of Queuing Systems.

Building Valid and Credible Simulation Models: Validation Principles and Techniques, Statistical Procedures for Comparing Real-World Observations and Simulated Outputs, Input Modeling; Generating Random Numbers and Random Variates; Output Analysis. Simulation Languages; Analysis and Modeling of Some Practical Systems

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding simulation modeling basics	✓											
CLO2	Understanding concepts in discrete-event simulation	✓											
CLO3	Creating building valid and credible simulation models			✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO3	Cognitive (Create)	• Lecture • Discussion • Interaction	• Class Test • Final Exam

Course Code: CSE 4116

Course Title: Simulation and Modeling Sessional

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: In this course, theoretical knowledge of the course CSE 4113 is implemented in the laboratory. Lab experiments are conducted according to the theory course.

Course Contents:

Sessional on Simulation and Modeling: Designing Discrete-event and continuous simulation model, Designing combined discrete-continuous model, Designing Monte Carlo simulation and simulation of queuing systems.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Designing Discrete-event and continuous simulation model.			✓		✓							
CLO2	Designing combined discrete-continuous model			✓		✓							
CLO3	Designing Monte Carlo simulation and simulation of queuing systems			✓		✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Create)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Performance • Project • Lab Final
CLO2	Cognitive (Create)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Performance • Project • Lab Final
CLO3	Cognitive (Create)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Performance • Project • Lab Final

Course Code: CSE 4117
Course Title: Wireless Networks

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: Course covers the fundamentals of wireless networks, emphasizing existing and future wireless networks. Some topics addressed will be cellular communication foundations, mobile radio propagation, multiple access strategies, mobility support and channel allocation. Other topics covered will be wireless ad-hoc networks, wireless sensor networks, and routing in wireless and mobile networks. This course introduces students to current wireless network technologies and topologies. We will introduce students to wireless networking research and help them conduct semester-long research projects. We will also examine recent industry developments and some new revolutionary ideas. The course will use research papers, industry white papers, and Internet RFCs as sources.

Course Contents:

Introduction to Wireless Networks: Wireless Access Networks, Wireless Mesh Networks, Personal Area Networks (Wireless Sensor Networks, Body Area Networks, Lowpan, and Bluetooth), Wireless and Mobile Ad Hoc Networks, Challenged Networks (Dtns, Vanets).

Wireless MAC Protocols: IEEE 802.11, IEEE 802.11e, IEEE 802.11n, IEEE 802.11s, IEEE 802.15.4, S-MAC, B-MAC, IEEE 802.22/20, IEEE 802.16d/e.

Wireless Routing: Routing Matrix ETX, ETT, WCETT, Air Time Metric, Routing Protocols AODV, DSR, DSDV, HWMP, Sensor Network Routing, VANET Routing etc.

Others: Wireless Transport Protocols; Wireless TCP and its Variants, Hop by Hop Congestion Control, Rate based Congestion Control etc. Quality of Service in Wireless Networks.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding wireless networks	✓											
CLO2	Analyze wireless MAC protocols		✓										
CLO3	Apply wireless routing algorithms				✓								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Analyze)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO3	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam

Course Code: CSE 4118

Course Title: Wireless Networks Sessional

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: Lab covers the simulated implementation of the fundamentals of wireless networks, emphasizing existing and future wireless networks. Some topics addressed will be cellular communication foundations, mobile radio propagation, multiple access strategies, mobility support and channel allocation. Other topics covered will be wireless ad-hoc networks, wireless sensor networks, and routing in wireless and mobile networks. This lab introduces students to current wireless network technologies and topologies. The lab will use research papers, industry white papers, and Internet RFCs as sources.

Course Contents:

Sessional on Wireless Networks: Introduction to Wireless Networks, Wireless MAC Protocols and routing.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	To study the communication networks characteristics		✓										
CLO2	To analyze various MAC and routing layer Protocols			✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO2	Cognitive (Apply)	• Lecture • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final

Course Code: CSE 4119
Course Title: Data Mining

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: Data mining techniques are the most important and useful tool for extracting novel, interesting and valid patterns and knowledge from a large amount of data. The objective of data mining processes is to build an efficient descriptive and predictive model of a large data set which will not only explain it but will also be able to generalize to new data sets.

Course Contents:

Introduction and Applications: Relational Databases, Data Warehouses, Transactional Databases, Advanced Data and Information Systems, Characterization and Discrimination, Classification and Prediction, Cluster Analysis, Outlier Analysis, Evolution Analysis.

Data Preprocessing: Descriptive Data Summarization, Data Cleaning, Data Integration and Transformation, Data Reduction, Data Discretization and Concept Hierarchy Generation.

Mining Frequent Patterns, Associations, and Correlations: Definitions and background, applications, closed and maximal frequent itemsets, Apriori and FP-growth algorithms, generating association rules from frequent itemsets, mining correlations, and different correlation measures.

Classification, Clustering and Prediction: Classification by Decision Tree Induction, Bayesian Classification, Rule-Based Classification, Clustering by Partitioning/ Hierarchical/ Density-based/ Model-based methods, Clustering High-Dimensional Data, Outlier analysis, Prediction, Linear Regression, Nonlinear Regression, Other Regression-Based Methods of prediction, Evaluating the Accuracy and error measures of a Classifier or Predictor.

Advanced Analysis: Mining Stream, Time-Series, and Sequence Data, Graph Mining, Social Network Analysis, and Multi-relational data mining, Mining Object, Spatial, Multimedia, and Text Data.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding the basic knowledge of data mining and its applications in computer science and engineering field	✓											
CLO2	Applying different data preprocessing techniques for different data sets					✓							
CLO3	Designing and implementing classification, clustering and prediction methods			✓									
CLO4	Explaining and Analyzing web mining and advanced data mining approaches		✓										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO3	Cognitive (Create)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO4	Cognitive (Understand, Analyze)	• Lecture • Discussion • Interaction	• Class Test • Final Exam

Course Code: CSE 4120

Course Title: Data Mining Sessional

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: Data mining techniques are the most important and useful tool for extracting novel, interesting and valid patterns and knowledge from a large amount of data. The objective of data mining processes is to build an efficient descriptive and predictive model of a large data set which will not only explain it but will also be able to generalize to new data sets.

Course Contents:

Sessional on Data Mining: Introduction, Data Preprocessing, Mining Frequent Patterns, Associations, and Correlations, Classification, Clustering and Prediction, Advanced Analysis.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Applying the basic knowledge of data mining and its applications in computer science and engineering field					✓							
CLO2	Applying different data preprocessing techniques for different data sets					✓							
CLO3	Designing and implementing classification, clustering and prediction methods			✓									
CLO4	Explaining and Analyzing web mining and advanced data mining approaches		✓										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Apply)	• Lecture • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO2	Cognitive (Apply)	• Lecture • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO3	Cognitive (Create)	• Lecture • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO4	Cognitive (Understand, Analyze)	• Lecture • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final

Course Code: CSE 4121
Course Title: Computer Vision

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: The course will provide an overview of the challenges of vision, the common approaches and current techniques. In this course students will learn basic principles of image formation, image processing algorithms and different algorithms for 3D reconstruction and recognition. This course emphasizes the core vision tasks of scene understanding and recognition.

Course Contents:

Introduction and Image Formation: Background, Human vision, Computer Vision and Computer Graphic, Geometry, Photometry (brightness and color), Quantization, Camera calibration, Photometric Stereo, Depth from Defocus, Construction of 3D models.

Image segmentation and Feature Extraction: Image preprocessing, Image representations (continuous and discrete), Convolution, Filtering, Edge detection, Contour based representation, Region based representation, Deformable curves and surfaces, Snakes and active contours, Level set representations, Fourier and wavelet descriptors, Medial representations, Multiresolution analysis, SIFT features.

Motion Estimation: Regularization theory, Optical computation, Stereo Vision, Motion Estimation, Structure from motion, Shape from stereo and motion, Feature matching, Surface fitting, Active ranging.

Object Recognition: Detectors and Descriptors, SIFT & Single Object Recognition, Optical Flow & Tracking, HoG/SIFT features, Hough transforms and other simple object recognition methods, Principal component analysis, Bayes classifiers, SVM classifiers, Human Motion Recognition.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding the foundation of image formation and image analysis.	✓											
CLO2	Identifying basic concepts, terminology, theories, models and methods in the field of computer vision.				✓								
CLO3	Analyze and design a range of algorithms for image processing and computer vision.		✓										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Analyze)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Presentation
CLO3	Cognitive (Analyze)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Project

Course Code: CSE 4122

Course Title: Computer Vision Sessional

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: The course will provide an overview of the challenges of vision, the common approaches and current techniques. In this course students will learn basic principles of image formation, image processing algorithms and different algorithms for 3D reconstruction and recognition. This course emphasizes the core vision tasks of scene understanding and recognition.

Course Contents:

Sessional on Computer Vision: Introduction and Image Formation, Image segmentation and Feature Extraction, Motion Estimation, Object Recognition.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CO1	Understanding the foundation of image formation and image analysis.	✓											
CO2	Identifying basic concepts, terminology, theories, models and methods in the field of computer vision.				✓								
CO3	Analyze and design a range of algorithms for image processing and computer vision.		✓										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO2	Cognitive (Analyze)	• Lecture • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO3	Cognitive (Analyze)	• Lecture • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final

Course Code: CSE 4123

Course Title: Knowledge Engineering

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: This course is designed to give undergraduate level fundamental techniques applied by knowledge engineers to build intelligent systems: expert systems, knowledge-based systems, knowledge-based decision support systems, expert database systems, etc. In Knowledge Engineering students learn to collect and organize valuable data using mathematics and intelligent computer techniques. Students also use models and computers for data analysis, generating new knowledge and conclusions to solve practical problems and make decisions quickly and efficiently. Knowledge engineering is also related to mathematical logic, as well as strongly involved in cognitive science and socio-cognitive engineering where the knowledge is produced by

socio-cognitive aggregates and is structured according to our understanding of how human reasoning and logic works.

Course Contents:

Introduction: Fundamental of knowledge-based systems or knowledge-based decision support systems and expert database systems, Classical mathematical models vs. knowledge representations (KRs), Relational KR, Logical KR, Automated knowledge extraction from large data sets, Knowledge discovery in databases, Attributes' domains discretization and other data mining problems, Knowledge Acquisition and Sharing, Sharing Ontologies, Language Ontologies, Language Patterns, Tools for Knowledge Acquisition.

Knowledge representation - Using Predicate logic - representing facts in logic, functions and predicates, Conversion to clause form, Resolution in propositional logic, Resolution in predicate logic, Unification. Representing Knowledge Using Rules: Procedural Versus Declarative knowledge, Logic Programming, Forward versus Backward Reasoning.

Reasoning: Proportional Case, Handling Variables and Qualifies, Dealing with Intractability, Reasoning with Horn Clauses, Procedural Control of Reasoning, Rules in Production, Description Logic, Vivid Knowledge, Beyond Vivid, Closed World Reasoning, Non categorical Reasoning, Vagueness, Uncertainty and Degrees of Belief, Objective and Subjective Probability.

Learning: What is learning, Rote learning, Learning by Taking Advice, Learning in Problem-solving, Learning from example: induction, Explanation-based learning.

Connectionist Models: Hopfield Networks, Learning in Neural Networks, Applications of Neural Networks, Recurrent Networks. Connectionist AI and Symbolic AI, Fuzzy Logic.

Expert System: Representing and using Domain Knowledge, Reasoning with knowledge, Expert System Shells, Support for explanation- examples –Knowledge acquisition-examples.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Acquiring understanding of issues related to using computers in solving engineering problems concerning knowledge.	✓											
CLO2	Developing skills in formulating mathematical models based on available knowledge or data, in designing solution algorithms to analysis and decision problems, and in application of existing software tools		✓										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	- Lecture - Discussion - Interaction	- Class Test - Final Exam - Assignment

CLO2	Affective (Apply)	• Lecture • Tutorial • Discussion • Interaction	• Class Test • Final Exam • Presentation
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Course Code: CSE 4124

Course Title: Knowledge Engineering Sessional

Contact Hours/Week: 3/2 Hours

Credit Hour: 0.75

Rationale: This course is designed to give undergraduate level fundamental techniques applied by knowledge engineers to build intelligent systems: expert systems, knowledge-based systems, knowledge-based decision support systems, expert database systems, etc. In Knowledge Engineering students learn to collect and organize valuable data using mathematics and intelligent computer techniques. Students also use models and computers for data analysis, generating new knowledge and conclusions to solve practical problems and make decisions quickly and efficiently. Knowledge engineering is also related to mathematical logic, as well as strongly involved in cognitive science and socio-cognitive engineering where the knowledge is produced by socio-cognitive aggregates and is structured according to our understanding of how human reasoning and logic works.

Course Contents:

Sessional on Knowledge Engineering: Knowledge representation, Reasoning, Learning, Connectionist Models, Expert System

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Acquiring understanding of issues related to using computers in solving engineering problems concerning knowledge.	✓											
CLO2	Developing skills in formulating mathematical models based on available knowledge or data, in designing solution algorithms to analysis and decision problems, and in application of existing software tools		✓										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO2	Affective (Apply)	• Lecture • Discussion	• Quiz • Viva

		• Interaction • Audio/Video • Lab Manual	• Lab Report • Project • Lab Final
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4th YEAR 2nd SEMESTER

Course Code: CSE 4000
Course Title: Project/ Thesis II

Contact Hours/Week: 6 Hours
Credit Hour: 3.00

Rationale: A thesis is a document submitted in support of candidature for an academic degree or professional qualification presenting the students' research and findings. On the other hand, a project is any undertaking, carried out individually or collaboratively and possibly involving research or design, that is carefully planned to achieve a particular aim. In this course, students can choose any one of these two.

Course Contents:

Continuation of Project/Thesis Topic Under Taken in CSE 4000.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO 1	Explore, Choose and Build, create or design a solution for a real-life problem.		✓	✓									
CLO 2	Investigate the existing solutions on the chosen real-life problem and explore their strengths and limitations.				✓								
CLO 3	Explore and utilize modern tools to design the solution.					✓							
CLO 4	Utilize the complex engineering knowledge & activities to find a solution that will be beneficial for the society.						✓						
CLO 5	Design the solution model keeping environment and sustainability in mind.							✓					
CLO 6	Learn and exercise work and publications ethics while publishing the proposed solution(s).								✓				

CLO 7	Work both individually and in team to nurture personal skills and leadership.									✓			
CLO 8	Learn to communicate and present the proposed methodologies and solutions in an interactive manner.										✓		
CLO 9	Analyze the financial aspects and manage the project/thesis in a manner so that the solution can be kept feasible.											✓	
CLO 10	Utilize the designed solutions and publications to create career opportunities.												✓

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Create)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CLO2	Cognitive (Evaluate)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CLO3	Cognitive (Apply)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CLO4	Affective (Internalize)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CLO5	Psychomotor (Articulate)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CLO6	Affective (Value)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense

CLO7	Affective (Organize)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CLO8	Affective (Respond)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CLO9	Cognitive (Evaluate)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book • Defense
CLO10	Psychomotor (Embody)	• Tutorial • Discussion • Interaction • Audio/Video	• Cont. Asses. • Book

Course Code: CSE 4201

Course Title: Computer Graphics

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: This course motivates to enhance the ability in order to rapidly visualize, design and modify different types of shapes, structures and images interactively that is absolutely mandatory in the field of engineering and imaging technology.

Course Contents:

Introduction: Applications of Computer Graphics, Graphical User Interface, Graphics Hardware Display Devices, Vector Graphics and Raster Graphics System, Graphics Pipeline, Computer Animation.

Scan Conversion Algorithms: Midpoint Line, Circle and Ellipse Creation Algorithms, Slope Independent Line Drawing using Midpoint Line Algorithm.

Polygons: Standard Graphics Primitives, Difference Type of Polygons, Polygon Filling, Triangulation, Polygon Filling Algorithm.

Transformation: Homogeneous Coordination, Transformation in 3D, Transformation Matrices, Translation, Rotation, Scaling.

Windowing and Clipping: Window Viewpoint, Zooming, Panning, Line, Text and Polygon, Clipping Algorithms.

Projection: Parallel and Perspective, Standard Projection Matrices.

Hidden Surface Removal: Z-Buffering, Ray Casting Methods, BSP Trees Method, Back-Face Detection.

Curves and Surfaces: Polygonal surfaces, Parametric Cubic Curves- Hermite, Bezier and B-spline curves, Parametric Bicubic Surfaces: Bicubic splines.

Illumination and Shading: Light Models, Ambient light, Diffuse and Specular Reflection, Light Attenuations, Goraud and Phong Shading, Recursive Ray Tracing.

Monochrome and Colored Light: Monochrome Light, Additive and Subtractive Light, Colored Light- RGB, CMY, YIQ, HSV and HLS Color Model, Image File Format: PPM file, BMP file.

API Specifies: Introduction to Graphics Programming using Modern OpenGL (3.3+), Drawing Shapes with Modern OpenGL, Graphics Pipeline of Modern OpenGL.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding computer graphics fundamentals and modern opengl basics.	✓											
CLO2	Understanding basic graphics algorithms like scan conversion algorithms and polygons.	✓											
CLO3	Apply transformation, windowing and clipping and project in graphics and their application using modern opengl.		✓	✓		✓							
CLO4	Understanding hidden surface problems and curves & surfaces in graphics.	✓	✓										
CLO5	Understanding illumination & shading and monochrome & colored light and their application using modern opengl.	✓	✓	✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO3	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO4	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO5	Cognitive (Understand)	• Lecture	• Class Test

		• Discussion • Interaction	• Final Exam
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Course Code: CSE 4202

Course Title: Computer Graphics Sessional

Contact Hours/Week: 3/2Hours

Credit Hour: 0.75

Rationale: In this course, the theoretical knowledge is implemented in the laboratory-based on the course CSE-4201 (Computer Graphics and Animations). Lab experiments are outlined according to the theory course. At the end of this course, students can obtain a basic concept of OpenGL graphics library; experience the practical implementation of modeling, rendering & viewing of objects in 2D & 3D; become familiar with simple animation, lighting, coloring & texturing; and design 2D and 3D mini-projects.

Course Contents:

Sessional on Computer Graphics: Introduction to Modern OpenGL, OpenGL Transformations and Camera, OpenGL Lighting, Writing Simple Graphics Program.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO 1	Implementing basic graphics building blocks.		✓	✓						✓			
CLO 2	Understanding modern OpenGL fundamentals.	✓				✓							
CLO 3	Applying transformations and camera in modern OpenGL.		✓	✓		✓							
CLO 4	Applying shader, textures and Lighting in modern OpenGL.		✓	✓		✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO2	Cognitive (Create)	• Lecture • Discussion • Interaction	• Quiz • Viva • Lab Report • Project

		• Lab Manual	• Lab Final
CLO3	Cognitive (Create)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO4	Cognitive (Understand)	• Lecture • Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final

Course Code: CSE 4203

Course Title: Machine Learning

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: This course aims at introducing the fundamental theory and concepts of Machine Learning algorithms in particular Supervised, Unsupervised and Ensemble Learning and their applications. This will help students to get sufficient knowledge to analyze and design the various machine learning algorithms and apply the algorithms in real life problems and practical deployment of the models.

Course Contents:

Introduction: Introduction to Concepts of Supervised learning, Unsupervised Learning, Reinforcement Learning, Feature, Hypothesis, Prediction Model, Loss Function, Evaluation Metrics, Optimization and Brief Review of Required Mathematics.

Classification and regression: Nearest Neighbor algorithm, Linear Discriminant Analysis (LDA), Linear Regression, Logistic Regression.

Artificial Neural Networks: Perceptron Learning Rule, Multilayer Perceptron, Backpropagation, Backpropagation with time, Activation Functions, Loss Functions, Gradient Descent Optimization, Adam, Rmsprop, Bias Variance and Error analysis, Regularization (L1, L2, Dropout), Kohonen Self-organizing Maps.

Decision Tree Learning: ID3, CART, Overfitting Issue, Incorporating Continuous-Valued Attributes, Measures for Selecting Attributes, Handling Training Examples with Missing Attribute Values.

Bayesian Learning: Naïve Bayes classifier, Bayesian Networks.

Support Vector Machine: Maximum Margin Classification, Soft Margin Optimization, Gradient Ascent, Sequential Minimal Optimization, Support Vector Machine for Binary and Multiclass Classification, Hyperparameter Tuning.

Clustering: K-means, K-medoids, Density-based Hierarchical Clustering, Mixture of Gaussians, Expectation Maximization.

Ensemble Methods: Bagging, Boosting: Adaboost, Random Forests.

Application: Student should develop a predictive modeling application on imaging, natural language or IoT data.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding fundamentals of machine learning and its applications.		✓		✓	✓							
CLO2	Understanding and implementing classifiers, regression and decision trees.		✓	✓	✓								
CLO3	Understanding and implementing artificial neural networks		✓	✓	✓								
CLO4	Understanding and implementing Bayesian learning and support vector machine		✓	✓	✓								
CLO5	Understanding and applying clustering and ensemble methods		✓	✓	✓								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Evaluation)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO2	Cognitive (Analysis)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO3	Cognitive (Analysis)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO4	Cognitive (Synthesis)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Project
CLO5	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Project

Rationale: This course aims at introducing the fundamental theory and concepts of Machine Learning algorithms in particular Supervised, Unsupervised and Ensemble Learning and their applications. This will help students to get sufficient knowledge to analyze and design the various machine learning algorithms and apply the algorithms in real life problems and practical deployment of the models.

Course Contents:

Sessional on Machine Learning: Introduction, Classification and Regression, Artificial Neural Networks, Decision Tree Learning, Bayesian Learning, Support Vector Machine, Clustering, Ensemble Methods, Application.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Develop decision tree, classification & regression, bayesian, SVM and ensemble machine learning algorithms.			✓									
CLO2	Apply artificial neural networks machine learning algorithms.		✓										
CLO3	Apply clustering algorithms.			✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Create)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO2	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO3	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final

Course Code: CSE 4205
Course Title: Security and Privacy

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: Network security has become one of the major concerns that involves any measure taken to protect the integrity and usability of network and data. Students who are Computer Science and Engineering major need to enhance the skills of providing network security to protect the shared data from unauthorized access, interference and misuse. This course is designed to give undergraduate level fundamental knowledge in network security.

Course Contents:

Security Fundamentals: Confidentiality, Integrity and Availability, Assurance, Authenticity and Anonymity, Threats and Attacks, Security Principles.

Privacy: Why Does Privacy Matter, Defining Privacy, Protecting Privacy, Privacy Threats, Privacy in Media, Surveillance and Data, Privacy Laws.

Cryptography: Encryption and Decryption, Substitution Cipher, Transposition Cipher, One Time Pads, Symmetric Cryptography, Public Key Cryptography, Authentication Protocol 1.0 to 5.0, Cryptographic Hash Functions, Digital Signature, Digital Certificates, Key Distribution Center, Different Symmetric Key Algorithm, Certificate Authority, Security Standards: DES, RSA, Digital Signature Algorithm (DSA), SHA, AES.

Malicious Software: Types of Malware, Infected Content, Vulnerability Exploits, Social Engineering, System Corruption, Bots, Zombie, Key Loggers, Phishing, Viruses, Spyware, Backdoors, Software Security Issues, Counter Measures, Forensics Analysis.

Network Security: Secure Socket Layer (SSL), Transport Layer Security (TLS), IPSec, AAA Standards, E-Mail Securities, PGP, S/MIME, Denial of Service (DOS) Attack, Eavesdropping, IP Spoofing, Firewall, Intrusion Detection and Prevention.

Other Security: Operating System Security, Database Security, Software Security, Web Security, Physical Security, Security for E-Commerce, Security for LAN and WAN, Penetration Testing.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

[illegible]

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	- Lecture - Discussion - Interaction	- Class Test - Final Exam
CLO2	Cognitive (Analyze)	- Lecture - Discussion - Interaction Lab Manual	- Class Test - Final Exam - Project
CLO3	Cognitive (Understand)	- Lecture - Discussion - Interaction	- Class Test - Final Exam
CLO4	Cognitive (Apply)	- Lecture - Discussion - Interaction	- Class Test - Final Exam - Assignment
CLO5	Cognitive (Apply)	- Lecture - Tutorial - Discussion	- Class Test - Final Exam - Presentation

Course Code: CSE 4206**Course Title: Security and Privacy Sessional****Contact Hours/Week: 3/2 Hours****Credit Hour: 0.75**

Rationale: This course is designed to develop practical knowledge on security and privacy. Students will implement several cryptographic algorithms. Besides, penetration testing and the concept of ethical hacking in a real-world system will be introduced through this course.

Course Contents:

Sessional on Security and Privacy: Fundamentals, Cryptography, Malicious Software, Network Security, Penetration Testing.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding the concept of penetration testing and applying in different cases	✓				✓							
CLO2	Implementing different cryptography algorithms	✓				✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand & Apply)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final
CLO2	Cognitive (Apply)	• Discussion • Interaction • Lab Manual	• Quiz • Viva • Lab Report • Project • Lab Final

Course Code: CSE 4208
Course Title: Seminar

Contact Hours/Week: 3/2 Hours
Credit Hour: 0.75

Rationale: This course is designed to provide knowledge about different research areas, research papers, research ethics, analysis of research papers, preparation of the review papers and the presentation of final review papers before the audience.

Course Contents:

Students will work in groups or individually to prepare review papers on topics assigned by the teachers and will present before the audience.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Analyzing and preparing the review papers.				✓								
CLO2	Presenting the final review papers before the audience.	✓											

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Analyze)	• Discussion • Interaction	• Viva • Presentation
CLO2	Psychomotor (Embody)	• Discussion	• Viva

CLO4	Have an understanding of the characteristics of MOS circuit construction and the comparison between different state-of-the-art MOS technologies and processes.			✓									
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Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Apply & Create)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO3	Cognitive (Analyze)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO4	Cognitive (Create)	• Lecture • Discussion • Interaction	• Class Test • Final Exam

Course Code: CSE 4211

Course Title: Parallel and Distributed Processing

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: The primary objective of this course is to introduce the fundamentals of parallel and distributed processing, including system architecture, programming model, and performance analysis. It will focus on the basic architectural, programming, and algorithmic concepts in the design and implementation of parallel and distributed applications. The specific topics include, but not limited to, MPI, openMP, middleware, SMP, clusters, peer-to-peer, CUDA programming.

Course Contents:

Parallel Processing: Multiple Processor Organizations, Symmetric Multiprocessors, Cache Coherence, Multithreading and Chip Multiprocessors, Clusters, Hybrid, Grid, NUMA, Cloud Computing, Multicore Computers, General-Purpose GPU, Performance Measures: Amdahl's law.

Parallel Programming: Motivation for Parallelism, Concurrency and Synchronization, Data and Work Partitioning, Common Parallelization Strategies, Granularity, Load Balancing, Shared-Memory Programming: pthreads, OpenMP, Distributed-Memory Programming: MPI, GPU Programming: CUDA, OpenCL. Examples: Parallel Search, Parallel Sorting, etc., MapReduce.

Distributed Processing: System Architecture, Communication, Mid-session Recess, Replication & Consistency, Distributed Transactions, Distributed Shared Memory, Synchronization & Coordination, Concurrency issues, Middleware, Fault Tolerance, Security, Naming, Distributed File Systems.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understand working of parallel processing systems.	✓											
CLO2	Apply parallel programming to write efficient programs.		✓	✓									
CLO3	Understand internals of distributed processing systems.	✓											

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO2	Cognitive (Apply)	• Lecture • Tutorial • Discussion	• Class Test • Final Exam • Assignment
CLO3	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction	• Class Test • Final Exam • Assignment • Presentation

Course Code: CSE 4213**Course Title: Impact of Computer on Society****Contact Hours/Week: 3 Hours****Credit Hour: 3.00**

Rationale: Computers have changed the way people relate to one another and their living environment, as well as how humans organize their work, their communities, and their time. Society, in turn, has influenced the development of computers through the needs people have for processing information.

Course Contents:

Introduction: History of Computer Hardware, Software, Networking; Overview of Technological Change; Impact of Information Technology on Some Sectors.

Privacy and Personal Information: Personal Privacy, Computer Technology Effect Privacy, Moral Problems from Privacy Violation, Privacy Protection: Ethical and Legal Basis, Technological Strategies, Implications of Database Systems.

Freedom of Expression in Cyberspace: Offensive Speech and Censorship in Cyberspace, Pornography, Anonymity, Spam.

Computer & Software Reliability: Liability in Hardware and Software Failure, Responsibility s. Liability vs. Accountability, Historical Software Risks (such as the Therac-25 Case).

Intellectual Property: Intellectual Property, Copyrights, Patents, and Trade Secrets, Software Piracy, Software Patents, Pirated Software, Use of Licensed Software.

Computer Crime: History and Examples of Computer Crime; “Cracking” (“Hacking”) and its Effects; Viruses, Worms, and Trojan Horses; Online Scams, Identity Theft; Moral Issues Related to These Crimes.

Computer and Work: Impact of Employment, Work Environment, Employee Monitoring, Health Issues.

Professional Ethics and Responsibilities: Ethics, Computer Ethics, Ethical Guidelines for Computer Professionals, Examine and Discuss Professional Codes of Ethics, Conduct, and Practice (IEEE, ACM, SE, AITP, and so forth).

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understand the impact of computer on society.	✓											
CLO2	Analyze freedom of expression in cyberspace based on computer and software reliability.		✓										
CLO3	Investigate professional ethics and responsibilities of computer and work.				✓								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Class Test • Final Exam • Assignment
CLO2	Cognitive (Analyze)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Class Test • Final Exam • Assignment
CLO3	Cognitive (Analyze)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Class Test • Final Exam • Assignment

Course Code: CSE 4215

Course Title: Decision Support System

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: This course is designed to give undergraduate level a basic understanding of characteristic, applications and capabilities of Decision Support Systems. It will enable the students to better understand the decision management process and thus make strategic decisions. This is a very important course for any student who wants to have a complete command over decision making in any kind of critical environment.

Course Contents:

Introduction to Decision Support System: DSS Characteristics, Applications of Decision Support Systems, Capabilities of Decision Support Systems, Components of Decision Support Systems, Benefits of using DSS Systems.

Making Decisions in the Decision Support Systems Environment: Activities in the Decision Support Systems Environment, the Decision-Making Process, Information Use for Strategic Management, Making Decisions in the Decision Support Systems.

Environment: Strategic Analysis for the Organization, Types of Problems in the Decision-Making Process.

Developing Decision Support System: Approaches to DSS Development, DSS Software Tools, DSS Hardware and Operating System Platforms, Building and Implementing Decision Support Systems. Decision Support Systems in Detail: Types of Decision Support Systems, DSS Models, Data Mining, Group Decision, Support Systems, Executive Information Systems (EIS). Artificial Intelligence and Expert Systems, Systems Integration and the Future of DSS: Brainstorming.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Have a clear Concept about fundamental components of DSS, capabilities and its advantages.	✓	✓										
CLO2	Develop a clear understanding of decision making process in DSS and the usage of information for strategic management.		✓	✓							✓		
CLO3	To gather knowledge about different types of DSS and other key factors related to DSS,				✓	✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Analyze)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Presentation
CLO3	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Project

Course Code: CSE 4217
Course Title: Network Planning

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: Network planning and design is an iterative process, encompassing topological design, network-synthesis, and network-realization, and is aimed at ensuring that a new telecommunications network or service meets the needs of the subscriber and operator. This course is designed to give undergraduate level of knowledge in network planning and design.

Course Contents:

Introduction: Network components, Theoretical Network, Real World Networks. Network Architectural Design: Designing the LAN, Configuring the Network Server and Client, Network Administration, Remote Access, Expanding the Network, Wide Area Network Troubleshooting, Major Protocol Suites.

Network Simulation: Network Simulation and Optimization, Network Operations, Control and Maintenance, Network Administration, Network Management Database and Tools, Capacity Planning.

Network Optimization: Network Security and Integrity, Linear Programming and Network Algorithms for Planning, Reliability Theory and Network Planning.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding the fundamental knowledge of different networks.	✓											
CLO2	Implementing the knowledge of network through simulation.			✓									
CLO3	Implementing various techniques in network optimization.			✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Apply)	• Lecture • Tutorial • Discussion • Interaction	• Class Test • Final Exam • Project
CLO3	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment • Presentation

Course Code: CSE 4219
Course Title: Human Computer Interaction

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: This course is designed to give undergraduate level fundamental knowledge in human computer interaction as well as the information processing systems and approaches to execute human computer interaction. It also enables the students to understand the stakeholder's requirements and communicate more effectively with them. This is a very important course for anyone who wants to get involved in management of product in any technological company.

Course Contents:

Process and Model: Introduction to Human-Computer Interaction (HCI), Human Information Processing Systems, Models of Interaction, Approaches to HCI, User Interface, HCI in Software Process, Cognitive Models.

Issues and Requirements: Socio-organizational Issues and Stakeholders Requirements, Communication and Collaboration Models, Task Analysis, Dialog Notation and Design, Groupware, CSCW and Social Issues.

User System Interaction: Analysis and Design, User Interface Design, Interface Technique and Technology, Case Studies

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Have a clear perception of different approaches of human computer Interaction and different models of Interaction and its uses in software processes.	✓	✓										
CLO2	Development of collaboration and communication for the better understanding of stakeholder's requirements.									✓			
CLO3	Creating User Interface using different interface technology based on Analysis and case study.					✓							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Affective (Organize)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Presentation
CLO3	Psychomotor (Embody)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Project

Course Code: CSE 4221
Course Title: Switching Systems

Contact Hours/Week: 3 Hours
Credit Hour: 3.00

Rationale: Switching is the process through which nodes in a network are connected. Information can be transmitted once a link has been established. Telephone switching is commonly used to refer to the switching of voice channels.

Course Contents:

Evolution of Switching Systems: The Role of Switching Systems in Telecommunication Networks, Step by Step and Crossbar, Stored Program Control (SPC), Digital Switching, ATM Switching.

Switching System Architecture: Subscriber and Line Interface, Switching Network: Matrix and Channel Graph Representations, Blocking, Non-Blocking, and Rearrangeable Networks, Control Unit, Operation and Maintenance, Switching Process: Call Detecting, Number Analysis, Call Routing, Supervision, and Metering, Signaling Equipment.

Hardware and Software Structure of the Digital Switch: Time Switches and Space Switches, Path Searching, Processor Systems Architecture and Functions, Reliability and Fault Recovery, Man Machine Interface (MMI), Examples of the Present Digital Switching Systems.

ATM Switching Architectures and Performance: ATM Switch Architectures, Full-, and Partial-Connection Multistage Networks, Self-Routing Networks, ATM Switching.

Structures: Minimum-Depth Blocking Networks, Non-Blocking Single-, and Multiple-Queuing Networks, Arbitrary-Depth Blocking Networks, Fault-Tolerant ATM Switching Architectures.

New Trends in Switching: Photonic Switching, IP Switching.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understand and learn different switching systems and their architectures.	✓											
CLO2	Analysis of different aspects of switching and their performances.		✓										
CLO3	Investigate structures and new trends in switching systems.				✓								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
CLO1	Cognitive (Understand)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Class Test • Final Exam • Assignment

CLO2	Cognitive (Analyze)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Class Test • Final Exam • Assignment
CLO3	Cognitive (Analyze)	• Lecture • Tutorial • Discussion • Interaction • Audio/Video	• Class Test • Final Exam • Assignment

Course Code: CSE 4223

Course Title: Control System Engineering

Contact Hours/Week: 3 Hours

Credit Hour: 3.00

Rationale: Control System Engineering is the engineering approach taken to understand how the process can be managed by automation devices and to implement such into operation. This course is designed to provide the undergraduate level of knowledge of the functionality of how devices monitor signals, control the process, how performance is measured including accumulation of statistical data, and how data is communicated to a host system or to peer control devices.

Course Contents:

Introduction to Control System: Conventional Control System, Steady State Response to Step, Ramp, and Parabolic Inputs, Transient Response, Poles and Zeros, Frequency Response from Pole-Zero Diagram, Routh's Stability Criterion; Block Diagrams, Canonical Forms, Transfer Functions and Signal Flow Graph, Root Locus, Frequency Response, Nyquist's Stability Criterion.

Modern Control System: Introduction, State Variable Analysis, Controllability and Observability, Application of Eigen Value, Linear Control System Design by State Feedback.

Controller Design: On-Off, Fuzzy, P, PI, PD and PID Types, Introduction to Programmable Logic Controllers (PLC), Temperature Control System, Position Control System.

Mapping Course Learning Outcomes (CLOs) with the Program Learning Outcomes (PLOs):

CLO No.	CLO Statement	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Understanding the fundamental knowledge of conventional control system.	✓											
CLO2	Having an introduction with the modern control system.		✓										
CLO3	Implementing various techniques in controller design.			✓									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLO No.	Domain/ level of Learning Taxonomy	Teaching-Learning Strategy	Assessment Strategy
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CLO1	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam
CLO2	Cognitive (Understand)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Assignment
CLO3	Cognitive (Apply)	• Lecture • Discussion • Interaction	• Class Test • Final Exam • Presentation

Proposed Program Curriculum

BACHELOR OF SCIENCE IN COMPUTER SCIENCE & ENGINEERING (CSE)

Part-D

(Effective from 2020-21 session)



Department of Computer Science & Engineering (CSE)
Rajshahi University of Engineering & Technology

20. Grading/Evaluation

1. Grading Scale and Grades

The letter grade system shall be used to assess the performance of the student and shall be as follows:

Range of Marks	Letter Grade	Grade Point
80 % or above	A+ (A Plus)	4.00
75% to less than 80 %	A (A Regular)	3.75
70 % to less than 75%	A- (A Minus)	3.50
65% to less than 70%	B+ (B Plus)	3.25
60% to less than 65%	B (B Regular)	3.00
55% to less than 60%	B- (B Minus)	2.75
50% to less than 55%	C+ (C Plus)	2.50
45% to less than 50%	C (C Regular)	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00
Incomplete	I	-

2. Absence during Semester

A student should not be absent from quizzes, tests etc. during the semester. Such absence will naturally lead to reduction in points/marks, which count towards the final grade. Absence in semester final examination due to lack of attendance (less than 50%) only of any courses will be considered as unregistered, hence requires a new registration with a regular semester.

A student who has been absent for short period, up to a maximum of three weeks due to illness, should approach the course teacher(s) or the course coordinator(s) for a make-up quizzes or assignments immediately on returning to the classes. Such request should be supported by medical certificate from University medical officer. The medical certificate issued by a registered medical practitioner (with the registration number shown explicitly or the certificates) will also be acceptable only in those cases where the student has valid reason for his/her absence from the University.

3. Grade Point Average (GPA) and Cumulative Grade Point Average (CGPA)

Grade point average (GPA) is the weighted average of the grade points obtained in all the courses passed/completed by a student in a semester. 'F' grades do not count for GPA calculation. GPA of a semester will be calculated as follows:

$$CGPA = \frac{\sum_{i=1}^n C_i G_i}{\sum_{i=1}^n C_i}$$

Where, n is the total number of courses passed by the student, C_i is the number of credits allotted to a particular course i and G_i is the grade point corresponding to the grade awarded for i -th course.

The overall or Cumulative Grade Point Average (CGPA) gives the cumulative performance of the student from first semester up to any other semester which it refers and is computed by dividing the total grade points ($\sum C_i G_i$) accumulated up to the date by the total credit hours ($\sum C_i$). Both GPA and CGPA are rounded off to the second place of decimal for reporting.

4. Withdrawal from a Semester

If a student is unable to complete the semester Final Examination due to illness, accident or any other valid reason etc., he/she may apply to the Head of the department. Each Department will decide for total withdrawal from the semester before the start of the semester final examination. He/she may choose not to withdraw any laboratory/sessional/design course if the grade obtained in such a course is 'D' or better. The application must be supported by a medical certificate from any authorized Medical Officer. The Academic Council will take the final decision about such applications. However, he/she will not be permitted to the next year class unless he/she completes the required credit for that year.

5. Incomplete (I) courses

A grade 'I' shall be awarded for the courses (like project & thesis, design etc.) in the 1st semester, which continue through to the even semester.

6. Retake

6.1 Backlog Examination

- i) There will be Backlog Examination after the publication of result of 2nd semester examination.
- ii) 'F' grade(s) obtained after semester examination will be considered as backlog course(s).
- iii) Students are allowed to sit for maximum 3 backlog courses in 1st and/or 2nd semester(s).
- iv) Class test marks of Backlog courses in 1st/2nd semester(s) will be counted for Backlog examination.
- v) Maximum B (B regular) grade will be counted in Backlog examination.

Backlog Courses: The course(s) which a student registered in a Semester but after Semester examination he/she obtained 'F' grade in that course(s).

6.2 Short Semester Examination

The Short Semester Examination on only backlog courses may be conducted for the students who have participated in their 4(four)/5(Five) year degree course (up to 4th /5th year backlog examination). A student can register maximum 5 (Five) incomplete courses including sessional, project and thesis to obtain Bachelor degree. The short semester examination will be arranged in a convenient time by the Head of the Department within 10 weeks of the publication of results of the final year backlog examination. The evaluation system will be the similar as regular semester. The students willing to appear at the short semester examination have to apply to the Head of the Department and with his permission must register within 15 (Fifteen) working days of publication of final year Backlog examination results. Maximum grade B will be counted in short semester examination.

7. Grade Improvement

The courses in which a student has obtained 'D' or a higher Grade will be counted as credits earned by him/her. Any course in which a student has obtained 'F' grade will not be counted towards his/her earned credits.

A student, who obtains an 'F' grade in any Core Course in any semester, he/she will have to repeat the course. If a student obtains an 'F' in an Optional Course, he/she may choose to repeat the course or take a substitute course if available.

'F' grades will be considered as backlog courses. 'F' grades will not be counted for GPA calculation but will stay permanently on the Grade Sheet and Transcript.

A student obtaining D grade in a course will be allowed to repeat the course for the purpose of grade improvement if CGPA of the student falls below 2.20. In such case he/she will be awarded the new grade thus he/she obtains or retains his/her previous grade if he/she fails.

8. Dropout

1. The names of the students shall be struck off and removed from the rolls on the following grounds:
 - i) Non-payment of University fees and dues within the prescribed period.
 - ii) Forced to discontinue his/her studies under disciplinary rules.
 - iii) Withdrawal of names from the rolls of the University on grounds acceptable to the Vice-Chancellor of the University/ nominated authority after having cleared all dues.
 - iv) Could not earn required credits for graduation as outlined in the respective curriculum and/or fulfill CGPA requirement within the maximum allowed time of 7 academic years. For Architecture maximum allowed time is 8 academic years.
2. Every student whose name has been struck off the rolls by exercise of the clauses (ii) of Article 1 seeking re-admission after expiry of the period for which he/she was forced to discontinue his/her studies, shall submit an application to the Head of the Department in the prescribed form before the commencement of the session to which he/she seeks re-admission. The Head of the Department shall forward the application to the Registrar of the University with his remarks. In case the readmission is allowed, the student will be required on payment of all dues to get him/her-self admitted no later than one week from the date of permission given by the Registrar. All readmission should preferably be completed before the session starts. The percentage of attendance of the re-admitted students shall be counted from the date of recommendation of the concerned Head of the department.
3. No student who has withdrawn his/her name under clause (iii) of Article 1 shall be given readmission.
4. In case, a student whose name has been struck off the rolls under clause (i) of Article 1 seeks readmission within the session in which his/her name was struck off, he/she shall be readmitted on payment of all the arrears fees and dues. But if he/she seeks readmission in any subsequent session, the procedure for his/her readmission will be the same as described under Article 2.
5. The application of a student for readmission will be considered if he/she applies within two academic sessions from the semester of discontinuance of his/her studies in the University. Other than debarment as punishment under the ordinance related to discipline, a student failing for any other reason whatsoever to become a candidate for a semester final examination in which he/she ought to have had in the usual process of his/her progressive academic activities, shall be considered to have discontinued his/her studies for the relevant semester together with striking the name off from current roll and two such discontinuance periods will be considered equivalent to that for one academic session. The maximum period of discontinuance under no circumstances is to exceed two academic sessions during a student's period of studies for the degree.
6. In case any application for readmission is rejected, the student may appeal to the Academic Council and, in this case, the decision of the Academic Council shall be final.
7. A student, whose name has been struck off the rolls by exercise of clause (iv) of Article 1, is not eligible to seek readmission.
8. After short semester, if any student fails to complete his/her required courses he/she will take readmission in the final year.